

Data Mining & Warehousing — Complete Math Workbook

Every numerical problem from the course, question then full solution, arranged serially by lecture.

PREPARED BY

Md Shahidul Islam Shabuz

COURSE TEACHER

Dr. Momotaz Begum, PhD

Professor & Head, Department of CSE, DUET

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Problems marked **2023 Qn** also appeared on the DUET 2023 paper; where the same mathematics appears twice in the source material it is solved once here, under its first occurrence.

Lecture 2.2 — Attribute Selection Measures in Decision-Tree Induction

All four problems use the same training set D : 14 tuples, class `buy_computer`, with 9 yes and 5 no.

RID	AGE	INCOME	STUDENT	CREDIT_RATING	BUY_COMPUTER
1	youth	high	no	fair	no
2	youth	high	no	excellent	no
3	middle-aged	high	no	fair	yes
4	senior	medium	no	fair	yes
5	senior	low	yes	fair	yes
6	senior	low	yes	excellent	no
7	middle-aged	low	yes	excellent	yes
8	youth	medium	no	fair	no
9	youth	low	yes	fair	yes
10	senior	medium	yes	fair	yes
11	youth	medium	yes	excellent	yes
12	middle-aged	medium	no	excellent	yes
13	middle-aged	high	yes	fair	yes
14	senior	medium	no	excellent	no

M1 Entropy / expected information $Info(D)$

Lecture 2.2

QUESTION

Compute the expected information (entropy) needed to classify a tuple in the training set D .

- 1 Write the formula. m is the number of classes, p_i the probability that a tuple belongs to class i .

$$Info(D) = - \sum_{i=1}^m p_i \log_2(p_i)$$

- 2 Here $m = 2$: 9 tuples are yes and 5 are no, out of 14.

$$Info(D) = - \left[\frac{9}{14} \log_2 \frac{9}{14} + \frac{5}{14} \log_2 \frac{5}{14} \right]$$

- 3 Convert to decimals. Use $\log_2 x = \ln x / \ln 2$ on the calculator.

$$= - \left[0.6429 \times (-0.6374) + 0.3571 \times (-1.4855) \right]$$

- 4 Multiply out and remove the leading minus.

$$= - \left[-0.4098 - 0.5304 \right] = 0.9403$$

Answer $Info(D) = 0.9403$ bits.

Shortcut A partition that is *pure* (all one class) has $Info = 0$; one that is *exactly balanced* has $Info = 1$. Write those down without any logarithm.

QUESTION

Using Information Gain, determine the splitting attribute at the root node of the decision tree for D .

- 1 Two formulas are needed — the weighted entropy after splitting on A , then the gain.

$$Info_A(D) = \sum_{j=1}^v \frac{|D_j|}{|D|} \times Info(D_j) \quad Gain(A) = Info(D) - Info_A(D)$$

Each partition's entropy is weighted by the *fraction of tuples* that fall into it.

- 2 **Attribute age** — three partitions. Count classes in each: youth 5 rows (2 yes, 3 no), middle-aged 4 rows (4 yes, 0 no), senior 5 rows (3 yes, 2 no).

$$\begin{aligned} Info_{youth} &= - \left[\frac{2}{5} \log_2 \frac{2}{5} + \frac{3}{5} \log_2 \frac{3}{5} \right] = -[0.4(-1.3219) + 0.6(-0.7370)] = 0.9710 \\ Info_{middle} &= - \left[\frac{4}{4} \log_2 \frac{4}{4} + 0 \right] = 0 \quad (\text{pure partition}) \\ Info_{senior} &= - \left[\frac{3}{5} \log_2 \frac{3}{5} + \frac{2}{5} \log_2 \frac{2}{5} \right] = 0.9710 \\ Info_{age}(D) &= \frac{5}{14}(0.9710) + \frac{4}{14}(0) + \frac{5}{14}(0.9710) = 0.3468 + 0 + 0.3468 = 0.6935 \\ Gain(age) &= 0.9403 - 0.6935 = \mathbf{0.2468} \end{aligned}$$

- 3 **Attribute student** — yes: 7 rows (6 yes, 1 no); no: 7 rows (3 yes, 4 no).

$$\begin{aligned} Info_{stu=yes} &= - \left[\frac{6}{7} \log_2 \frac{6}{7} + \frac{1}{7} \log_2 \frac{1}{7} \right] = -[0.8571(-0.2224) + 0.1429(-2.8074)] = 0.5918 \\ Info_{stu=no} &= - \left[\frac{3}{7} \log_2 \frac{3}{7} + \frac{4}{7} \log_2 \frac{4}{7} \right] = -[0.4286(-1.2224) + 0.5714(-0.8074)] = 0.9852 \\ Info_{student}(D) &= \frac{7}{14}(0.5918) + \frac{7}{14}(0.9852) = 0.2959 + 0.4926 = 0.7885 \\ Gain(student) &= 0.9403 - 0.7885 = \mathbf{0.1518} \end{aligned}$$

- 4 **Attribute credit_rating** — fair: 8 rows (6 yes, 2 no); excellent: 6 rows (3 yes, 3 no).

$$\begin{aligned} Info_{fair} &= - \left[\frac{6}{8} \log_2 \frac{6}{8} + \frac{2}{8} \log_2 \frac{2}{8} \right] = -[0.75(-0.4150) + 0.25(-2)] = 0.8113 \\ Info_{excellent} &= - \left[\frac{3}{6} \log_2 \frac{3}{6} + \frac{3}{6} \log_2 \frac{3}{6} \right] = 1.0000 \quad (\text{balanced}) \\ Info_{credit}(D) &= \frac{8}{14}(0.8113) + \frac{6}{14}(1.0000) = 0.4636 + 0.4286 = 0.8922 \\ Gain(credit_rating) &= 0.9403 - 0.8922 = \mathbf{0.0481} \end{aligned}$$

- 5 **Attribute income** — high: 4 rows (2 yes, 2 no); medium: 6 rows (4 yes, 2 no); low: 4 rows (3 yes, 1 no).

$$\begin{aligned} Info_{high} &= 1.0000 \quad (\text{balanced}) \\ Info_{medium} &= - \left[\frac{4}{6} \log_2 \frac{4}{6} + \frac{2}{6} \log_2 \frac{2}{6} \right] = -[0.6667(-0.5850) + 0.3333(-1.5850)] = 0.9183 \\ Info_{low} &= - \left[\frac{3}{4} \log_2 \frac{3}{4} + \frac{1}{4} \log_2 \frac{1}{4} \right] = 0.8113 \\ Info_{income}(D) &= \frac{4}{14}(1) + \frac{6}{14}(0.9183) + \frac{4}{14}(0.8113) = 0.2857 + 0.3936 + 0.2318 = 0.9111 \\ Gain(income) &= 0.9403 - 0.9111 = \mathbf{0.0292} \end{aligned}$$

- 6 Compare and pick the largest gain.

ATTRIBUTE	$Info_A(D)$	$Gain(A)$
age	0.6935	0.2468
student	0.7885	0.1518
credit_rating	0.8922	0.0481
income	0.9111	0.0292

Answer age has the highest gain (0.2468 bits), so **age is the splitting attribute at the root**. The middle-aged branch is already pure, so it becomes a leaf labelled yes.

QUESTION

Compute $GainRatio(income)$ for the `buy_computer` dataset, and comment on the value given in the lecture slide.

- 1 Information Gain favours attributes with many distinct values. C4.5 divides it by *split information*, which measures how widely the attribute splits the data.

$$SplitInfo_A(D) = - \sum_{j=1}^v \frac{|D_j|}{|D|} \times \log_2 \left(\frac{|D_j|}{|D|} \right)$$

- 2 Substitute the partition sizes $|D_{high}| = 4$, $|D_{medium}| = 6$, $|D_{low}| = 4$ out of 14. Note this uses only the *sizes*, never the class labels.

$$\begin{aligned} SplitInfo_{income}(D) &= - \left[\frac{4}{14} \log_2 \frac{4}{14} + \frac{6}{14} \log_2 \frac{6}{14} + \frac{4}{14} \log_2 \frac{4}{14} \right] \\ &= - [0.2857(-1.8074) + 0.4286(-1.2224) + 0.2857(-1.8074)] \\ &= - [-0.5164 - 0.5239 - 0.5164] = 1.5567 \end{aligned}$$

- 3 Divide the gain from M2 by this.

$$GainRatio(A) = \frac{Gain(A)}{SplitInfo_A(D)} \implies GainRatio(income) = \frac{0.0292}{1.5567} = 0.0188$$

Answer $SplitInfo_{income}(D) = 1.5567$ bits, $GainRatio(income) = 0.0188$.

M4 Gini Index (CART) and a binary split

QUESTION

Compute the Gini index of D , then evaluate the binary split $income \in \{low, medium\}$ versus $income \in \{high\}$.

- 1 Gini measures impurity as one minus the sum of squared class probabilities.

$$\begin{aligned} Gini(D) &= 1 - \sum_{i=1}^m p_i^2 \\ Gini(D) &= 1 - \left[\left(\frac{9}{14} \right)^2 + \left(\frac{5}{14} \right)^2 \right] = 1 - [0.4133 + 0.1275] = 0.4592 \end{aligned}$$

- 2 CART builds a strictly binary tree, so a k -valued attribute must be tested as $2^{k-1} - 1$ different binary splits. For $income$ ($k = 3$) that is $2^2 - 1 = 3$ splits. The weighted formula is

$$Gini_A(D) = \frac{|D_1|}{|D|} Gini(D_1) + \frac{|D_2|}{|D|} Gini(D_2)$$

- 3 Count the two partitions from the RID table. $D_1 = \{low, medium\}$: rows 4, 5, 6, 7, 8, 9, 10, 11, 12, 14 $\rightarrow$ 10 tuples, **7 yes, 3 no**. $D_2 = \{high\}$: rows 1, 2, 3, 13 $\rightarrow$ 4 tuples, **2 yes, 2 no**.

$$Gini(D_1) = 1 - [0.7^2 + 0.3^2] = 1 - [0.49 + 0.09] = 0.420$$

$$Gini(D_2) = 1 - [0.5^2 + 0.5^2] = 1 - [0.25 + 0.25] = 0.500$$

- 4 Weight by partition size.

$$Gini_{income \in \{low, medium\}}(D) = \frac{10}{14}(0.420) + \frac{4}{14}(0.500) = 0.3000 + 0.1429 = 0.4429$$

Answer $Gini(D) = 0.4592$; the split $\{low, medium\}$ vs $\{high\}$ gives $Gini = 0.4429$. CART chooses the split with the **lowest** Gini.

Sanity check For two classes, $Gini \leq 0.5$ and $Info \leq 1.0$. A Gini above 0.5 means an arithmetic slip.

Lecture 5 — Hierarchical Methods and Density-Based Methods

M5 AGNES and DIANA dendrogram

Lecture 5 · 2023 Q3(c)

QUESTION

Suppose there is a data set contains 10 objects $\{a, b, c, d, e, f, g, h, i, l, m, n, q, t\}$. Now design dendrogram according to hierarchical Clustering (Agglomerative and divisive algorithms).

Note on the question. It says “10 objects” but lists **14**. Answer for the 14 objects actually listed and state the discrepancy — that is what earns the mark.

- 1 Fix the direction of travel and the criterion for each algorithm.

	AGNES (AGGLOMERATIVE)	DIANA (DIVISIVE)
Strategy	Bottom-up	Top-down
Start	each object is its own cluster (14 clusters)	all 14 objects in one cluster
Each step	merge the two closest clusters	split one cluster into two
Criterion	minimum Euclidean distance between any two objects from different clusters (single linkage)	maximum Euclidean distance between the closest neighbouring objects in the cluster
Stop	one cluster remains	every object is alone

$$d_{min}(C_i, C_j) = \min_{p \in C_i, p' \in C_j} |p - p'| \quad d_{max}(C_i, C_j) = \max_{p \in C_i, p' \in C_j} |p - p'|$$

- 2 Count the levels. With n objects AGNES performs exactly $n - 1$ merges, so here

$$n = 14 \implies \text{number of merges} = 14 - 1 = 13$$

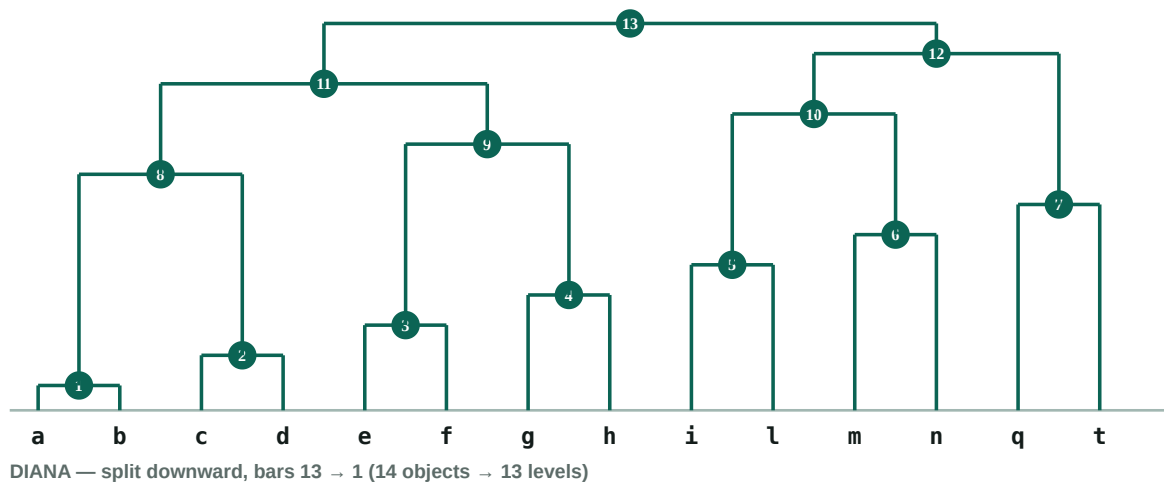
DIANA performs the same 13 operations as splits, in the reverse order. The dendrogram therefore has 13 horizontal bars.

- 3 The question supplies no coordinates or distance matrix, so state the assumption and proceed: *objects are merged in order of increasing pairwise distance, the nearest pair first*. The merge sequence below follows that rule.

STEP	MERGE (AGNES)	NEW CLUSTER	CLUSTERS REMAINING
1	a + b	(ab)	13
2	c + d	(cd)	12
3	e + f	(ef)	11
4	g + h	(gh)	10
5	i + l	(il)	9
6	m + n	(mn)	8
7	q + t	(qt)	7
8	(ab) + (cd)	(abcd)	6
9	(ef) + (gh)	(efgh)	5
10	(il) + (mn)	(ilmn)	4
11	(abcd) + (efgh)	(abcdefgh)	3
12	(ilmn) + (qt)	(ilmnqt)	2
13	(abcdefgh) + (ilmnqt)	(abcdefghilmnqt)	1

- 4 Draw the dendrogram. Each numbered bar is one merge; its height is the distance at which the merge occurred, so bars rise as the clusters get further apart.

AGNES — merge upward, bars 1 → 13



Dendrogram for the 14 objects. Read **upward** (bars 1 → 13) for AGNES; read **downward** (bars 13 → 1) for DIANA, where each bar becomes a split instead of a merge.

- 5 Read the two algorithms off the one diagram.

AGNES starts at the leaves and works up: $\{a\}, \{b\}, \dots, \{t\} \rightarrow (ab) \rightarrow (cd) \rightarrow \dots \rightarrow (abcdefghilmnqt)$.

DIANA starts at the root and works down: $(abcdefghilmnqt) \rightarrow (abcdefgh) + (ilmnqt) \rightarrow \dots \rightarrow 14 \text{ singleton clusters}$.

Cutting the dendrogram with a horizontal line at any height gives the clustering at that level — e.g. a cut just below bar 13 yields the **two** clusters (abcdefgh) and (ilmnqt).

Answer 14 objects require **13 merge levels**. AGNES merge order: ab, cd, ef, gh, il, mn, qt, (abcd), (efgh), (ilmn), (abcdefgh), (ilmnqt), (all). DIANA performs exactly the reverse sequence as splits.

Always add Hierarchical clustering has **no backtracking** — once a merge or split is made it can never be undone, even if it later proves poor. Also state: with no distance matrix supplied, the merge order is assumed from proximity.

M6 Clustering Feature and the additivity theorem

Lecture 5

QUESTION

Cluster 1 contains the three points $\{(2, 5), (3, 2), (4, 3)\}$ and Cluster 2 has $CF_2 = \langle 3, (35, 36), (417, 440) \rangle$. Compute CF_1 , and hence $CF_3 = CF_1 + CF_2$.

- 1 A Clustering Feature summarises a cluster in three numbers — the count, the linear sum and the square sum.

$$CF = \langle N, \overrightarrow{LS}, SS \rangle, \quad \overrightarrow{LS} = \sum_{i=1}^N \overrightarrow{X_i}, \quad SS = \sum_{i=1}^N \overrightarrow{X_i}^2$$

- 2 N_1 is just the number of points: $N_1 = 3$.

- 3 Linear sum — add the x values and the y values **separately**.

$$\overrightarrow{LS}_1 = (2 + 3 + 4, 5 + 2 + 3) = (9, 10)$$

- 4 Square sum — square each coordinate first, then add, again component by component.

$$SS_1 = (2^2 + 3^2 + 4^2, 5^2 + 2^2 + 3^2) = (4 + 9 + 16, 25 + 4 + 9) = (29, 38)$$

$$CF_1 = \langle 3, (9, 10), (29, 38) \rangle$$

- 5 Additivity theorem: for two **disjoint** clusters the merged CF is the component-wise sum. No raw data is needed — three additions.

$$CF_3 = CF_1 + CF_2 = \langle N_1 + N_2, \overrightarrow{LS}_1 + \overrightarrow{LS}_2, SS_1 + SS_2 \rangle$$

$$N_3 = 3 + 3 = 6 \quad \overrightarrow{LS}_3 = (9 + 35, 10 + 36) = (44, 46) \quad SS_3 = (29 + 417, 38 + 440) = (446, 478)$$

Answer $CF_1 = \langle 3, (9, 10), (29, 38) \rangle$ and $CF_3 = \langle 6, (44, 46), (446, 478) \rangle$.

Never collapse the pair SS for 2-D data is a *pair* $(\sum x_i^2, \sum y_i^2)$ — never the single number $\sum (x_i^2 + y_i^2)$. Losing the pair structure is the usual way to drop marks here.

M7 Clustering Feature of Cluster 3

Lecture 5 · 2023 Q4(a)

QUESTION

Calculate the cluster feature for Cluster 3, where Cluster 1 contains three objects $\{(2, 5), (3, 2), (4, 3)\}$ and Cluster 2 contains the objects $\{(4, 5), (3, 2), (3, 4)\}$.

- 1 CF_1 is the same as in M6: $CF_1 = \langle 3, (9, 10), (29, 38) \rangle$.

- 2 Compute CF_2 from its three points — this time the coordinates are given, so do not reuse the slide's CF_2 .

$$N_2 = 3$$

$$\overrightarrow{LS}_2 = (4 + 3 + 3, 5 + 2 + 4) = (10, 11)$$

$$SS_2 = (4^2 + 3^2 + 3^2, 5^2 + 2^2 + 4^2) = (16 + 9 + 9, 25 + 4 + 16) = (34, 45)$$

- 3 Cluster 3 is the union of the two, so apply additivity.

$$N_3 = 3 + 3 = 6 \quad \overrightarrow{LS}_3 = (9 + 10, 10 + 11) = (19, 21) \quad SS_3 = (29 + 34, 38 + 45) = (63, 83)$$

- 4 Optional extra marks — derive the centroid and radius of Cluster 3 straight from CF_3 .

$$\vec{x}_0 = \frac{\overrightarrow{LS}_3}{N_3} = \left(\frac{19}{6}, \frac{21}{6}\right) = (3.1667, 3.5000)$$

$$R = \sqrt{\left(\frac{63}{6} - 3.1667\right)^2 + \left(\frac{83}{6} - 3.5\right)^2} = \sqrt{0.4722 + 1.5833} = \sqrt{2.0556} = 1.4337$$

Answer $CF_3 = \langle 6, (19, 21), (63, 83) \rangle$.

Question wording The paper says Cluster 2 “contains **same** objects”, which is misleading — the coordinates differ from Cluster 1. Compute from the coordinates printed in the question, not from the slide's $CF_2 = \langle 3, (35, 36), (417, 440) \rangle$.

M8 Centroid, radius and diameter from a CF

Lecture 5

QUESTION

Using $CF_1 = \langle 3, (9, 10), (29, 38) \rangle$, compute the centroid, radius and diameter of Cluster 1 without touching the raw data points.

- 1 **Centroid** — the mean vector, i.e. the linear sum divided by the count.

$$\vec{x}_0 = \frac{\overrightarrow{LS}}{N} = \left(\frac{9}{3}, \frac{10}{3}\right) = (3, 3.33)$$

- 2 **Radius** — the average root-mean-square distance from the points to the centroid. The middle form is the one to use, because it needs only SS , LS and N .

$$R = \sqrt{\frac{\sum_{i=1}^n (\vec{X}_i - \vec{x}_0)^2}{n}} = \sqrt{\frac{SS}{N} - \left(\frac{\overrightarrow{LS}}{N}\right)^2}$$

- 3 Work out the variance in each dimension, then add and take the root.

$$x : \frac{29}{3} - 3^2 = \frac{29 - 27}{3} = \frac{2}{3} \quad y : \frac{38}{3} - \left(\frac{10}{3}\right)^2 = \frac{114 - 100}{9} = \frac{14}{9}$$

$$R = \sqrt{\frac{2}{3} + \frac{14}{9}} = \sqrt{\frac{6 + 14}{9}} = \sqrt{\frac{20}{9}} = \frac{\sqrt{20}}{3} \approx 1.49$$

- 4 **Diameter** — the root of the average squared distance over all pairs. The double sum is obtained from the CF by an identity, so again no raw points are needed.

$$\sum_{i=1}^N \sum_{j=1}^N (\vec{X}_i - \vec{X}_j)^2 = 2N \cdot SS - 2 \overrightarrow{LS}^2$$

$$x : 2(3)(29) - 2(9)^2 = 174 - 162 = 12 \quad y : 2(3)(38) - 2(10)^2 = 228 - 200 = 28$$

$$\text{total} = 12 + 28 = 40$$

- 5 Divide by the number of pairs and take the root. Two conventions exist — give both.

$$\text{standard BIRCH: } D = \sqrt{\frac{40}{N(N-1)}} = \sqrt{\frac{40}{6}} = \sqrt{\frac{20}{3}} \approx 2.58$$

$$\text{slide formula: } D = \sqrt{\frac{40}{n}} = \sqrt{\frac{40}{3}} \approx 3.65$$

Answer $\vec{x}_0 = (3, 3.33)$; $R \approx 1.49$; $D \approx 2.58$ (standard) or 3.65 (slide formula).

QUESTION

Compute the Jaccard coefficient for transaction items a, b, c, d, e, f, g forming two clusters of transactions:

Cluster 1 $\langle a, b, c, d, e \rangle$:

$\{a, b, c\}, \{a, b, d\}, \{a, b, e\}, \{a, c, d\}, \{a, c, e\}, \{a, d, e\}, \{b, c, d\}, \{b, c, e\}, \{b, d, e\}, \{c, d, e\}$

Cluster 2 $\langle a, b, f, g \rangle$: $\{a, b, f\}, \{a, b, g\}, \{a, f, g\}, \{b, f, g\}$

- 1 The Jaccard coefficient is the size of the intersection over the size of the union.

$$\text{sim}(T_i, T_j) = \frac{|T_i \cap T_j|}{|T_i \cup T_j|}$$

- 2 **Intra-cluster** — two transactions inside Cluster 1. Take $T_1 = \{a, b, c\}$ and $T_2 = \{b, d, e\}$.

$$T_1 \cap T_2 = \{b\} \Rightarrow |T_1 \cap T_2| = 1 \quad T_1 \cup T_2 = \{a, b, c, d, e\} \Rightarrow |T_1 \cup T_2| = 5$$

$$\text{sim}(\{a, b, c\}, \{b, d, e\}) = \frac{1}{5} = 0.2$$

- 3 **Inter-cluster** — one transaction from each cluster. Take $T_1 = \{a, b, c\}$ and $T_3 = \{a, b, f\}$.

$$T_1 \cap T_3 = \{a, b\} \Rightarrow 2 \quad T_1 \cup T_3 = \{a, b, c, f\} \Rightarrow 4$$

$$\text{sim}(\{a, b, c\}, \{a, b, f\}) = \frac{2}{4} = 0.5$$

- 4 Find the range inside Cluster 1. Two 3-item subsets of a 5-item set share either 1 or 2 items, so

$$\text{share 1: } \frac{1}{5} = 0.2 \quad \text{share 2: } \frac{2}{4} = 0.5$$

Similarity within Cluster 1 therefore ranges from **0.2 to 0.5**.

- 5 State the conclusion the examiner wants. Transactions from *different* clusters reach 0.5, which is **higher** than the 0.2 achieved by two transactions from the *same* cluster. A greedy merge driven by Jaccard would wrongly group $\{a, b, c\}$ with $\{a, b, f\}$ — so distance functions do not give high-quality clusters on categorical data. This motivates ROCK (M10).

Answer Intra-cluster $\text{sim} = 0.2$; inter-cluster $\text{sim} = 0.5$; within-Cluster-1 range 0.2 to 0.5.

Shortcut For two sets of size 3, $|T_i \cup T_j| = 6 - |T_i \cap T_j|$, so $\text{sim} = \frac{|\cap|}{6 - |\cap|}$. You never need to write the union out.

Question-paper typo The 2023 paper lists Cluster 1 with only **nine** subsets — $\{b, c, d\}$ is missing. There are $\binom{5}{3} = 10$. Note the omission and use all ten; M10 depends on the full set.

M10 ROCK — number of common neighbours (links)

QUESTION

For the same 14 transactions and $\theta = 0.5$, compute $\text{Link}(\{a, b, f\}, \{a, b, g\})$ and $\text{Link}(\{a, b, f\}, \{a, b, c\})$, and justify the statement “link is a better measure than the Jaccard coefficient”.

- 1 Two transactions are **neighbours** when their similarity reaches the threshold.

$$Sim(T_i, T_j) \geq \theta, \quad Sim(T_i, T_j) = \frac{|T_i \cap T_j|}{|T_i \cup T_j|}$$

For 3-item sets with $\theta = 0.5$, sharing 2 items gives $\frac{2}{4} = 0.5 \geq 0.5$. **Working rule: neighbours $\iff$ they share at least 2 letters.**

- 2 The link between two transactions is the number of *common* neighbours they have.

$$Link(T_i, T_j) = |N(T_i) \cap N(T_j)|$$

- 3 Build the neighbour list of each transaction involved, by scanning all 14 for a 2-letter overlap.

$$N(\{a, b, f\}) = \{ \{a, b, g\}, \{a, f, g\}, \{b, f, g\}, \{a, b, c\}, \{a, b, d\}, \{a, b, e\} \}$$

$$N(\{a, b, g\}) = \{ \{a, b, f\}, \{a, f, g\}, \{b, f, g\}, \{a, b, c\}, \{a, b, d\}, \{a, b, e\} \}$$

$$N(\{a, b, c\}) = \{ \{a, b, d\}, \{a, b, e\}, \{a, b, f\}, \{a, b, g\}, \{a, c, d\}, \{a, c, e\}, \{b, c, d\}, \{b, c, e\} \}$$

- 4 Intersect the lists. **Same cluster:**

$$N(\{a, b, f\}) \cap N(\{a, b, g\}) = \{ \{a, f, g\}, \{b, f, g\}, \{a, b, c\}, \{a, b, d\}, \{a, b, e\} \}$$

$$Link(\{a, b, f\}, \{a, b, g\}) = 5$$

- 5 **Different clusters:**

$$N(\{a, b, f\}) \cap N(\{a, b, c\}) = \{ \{a, b, d\}, \{a, b, e\}, \{a, b, g\} \}$$

$$Link(\{a, b, f\}, \{a, b, c\}) = 3$$

- 6 Justification. Direct Jaccard rates **both** pairs at 0.5 and so cannot decide which transaction belongs where. The link measure separates them: $5 > 3$, so $\{a, b, f\}$ is correctly grouped with $\{a, b, g\}$ in Cluster 2.

Answer $Link(\{a, b, f\}, \{a, b, g\}) = 5$, $Link(\{a, b, f\}, \{a, b, c\}) = 3$. Because $5 > 3$, ROCK places $\{a, b, f\}$ with $\{a, b, g\}$.

Model sentence “The Jaccard coefficient evaluates only isolated pairwise overlap and assigns misleadingly high similarity to transactions from different clusters. The link measure captures global neighbourhood context by counting shared mutual neighbours, preventing incorrect merges.”

M11 DBSCAN — 12-point classification and clustering

Lecture 5

QUESTION

Apply DBSCAN to the 12 points below with $\epsilon = 1.9$ and $MinPts = 4$. Compute all pairwise distances, classify every point as Core, Border or Noise, and report the final clusters.

POINT	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
x	3	4	5	6	7	6	7	8	9	2	3	2
y	7	6	5	4	3	2	2	4	5	6	5	4

- 1 Neighbours are decided by Euclidean distance.

$$d(P_i, P_j) = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2}$$

- 2 **Do not take square roots.** Compare squared values instead — this is the single biggest time saver on this question.

$$d(P_i, P_j) \leq \varepsilon \iff (x_j - x_i)^2 + (y_j - y_i)^2 \leq \varepsilon^2, \quad \varepsilon^2 = 1.9^2 = 3.61$$

So: sum the squared differences and check against **3.61**. Only sums of 1 or 2 can qualify — i.e. points one unit apart along an axis ($d = 1$) or one unit diagonally ($d = \sqrt{2} = 1.4142$). A sum of 4 gives $d = 2.0$, which already exceeds 1.9.

- 3 **Worked examples of the method,** before the full table.

$$d(P_1, P_2)^2 = (4 - 3)^2 + (6 - 7)^2 = 1 + 1 = 2 \leq 3.61 \Rightarrow \text{neighbour}, \quad d = \sqrt{2} = 1.4142$$

$$d(P_1, P_3)^2 = (5 - 3)^2 + (5 - 7)^2 = 4 + 4 = 8 > 3.61 \Rightarrow \text{not a neighbour}, \quad d = 2.8284$$

$$d(P_5, P_7)^2 = (7 - 7)^2 + (2 - 3)^2 = 0 + 1 = 1 \leq 3.61 \Rightarrow \text{neighbour}, \quad d = 1.0000$$

$$d(P_1, P_{11})^2 = (3 - 3)^2 + (5 - 7)^2 = 0 + 4 = 4 > 3.61 \Rightarrow \text{not a neighbour}, \quad d = 2.0000$$

That last one is the trap: $d = 2.0$ looks close but $2.0 > 1.9$, so P1 and P11 are **not** neighbours.

4 All $\binom{12}{2} = 66$ pairwise calculations. Neighbour pairs are highlighted; there are 13 of them.

PAIR	COORDS	($\Delta X, \Delta Y$)	$\Delta X^2 + \Delta Y^2$	D	≤ 3.61	PAIR	COORDS	($\Delta X, \Delta Y$)	$\Delta X^2 + \Delta Y^2$	D	≤ 3.61
P1– P2	(3, 7) (4, 6)	(1, -1)	$1+1 = 2$	1.4142	✓	P4– P8	(6, 4) (8, 4)	(2, 0)	$4+0 = 4$	2.0000	–
P1– P3	(3, 7) (5, 5)	(2, -2)	$4+4 = 8$	2.8284	–	P4– P9	(6, 4) (9, 5)	(3, 1)	$9+1 = 10$	3.1623	–
P1– P4	(3, 7) (6, 4)	(3, -3)	$9+9 = 18$	4.2426	–	P4– P10	(6, 4) (2, 6)	(-4, 2)	$16+4 = 20$	4.4721	–
P1– P5	(3, 7) (7, 3)	(4, -4)	$16+16 = 32$	5.6569	–	P4– P11	(6, 4) (3, 5)	(-3, 1)	$9+1 = 10$	3.1623	–
P1– P6	(3, 7) (6, 2)	(3, -5)	$9+25 = 34$	5.8310	–	P4– P12	(6, 4) (2, 4)	(-4, 0)	$16+0 = 16$	4.0000	–
P1– P7	(3, 7) (7, 2)	(4, -5)	$16+25 = 41$	6.4031	–	P5– P6	(7, 3) (6, 2)	(-1, -1)	$1+1 = 2$	1.4142	✓
P1– P8	(3, 7) (8, 4)	(5, -3)	$25+9 = 34$	5.8310	–	P5– P7	(7, 3) (7, 2)	(0, -1)	$0+1 = 1$	1.0000	✓
P1– P9	(3, 7) (9, 5)	(6, -2)	$36+4 = 40$	6.3246	–	P5– P8	(7, 3) (8, 4)	(1, 1)	$1+1 = 2$	1.4142	✓
P1– P10	(3, 7) (2, 6)	(-1, -1)	$1+1 = 2$	1.4142	✓	P5– P9	(7, 3) (9, 5)	(2, 2)	$4+4 = 8$	2.8284	–
P1– P11	(3, 7) (3, 5)	(0, -2)	$0+4 = 4$	2.0000	–	P5– P10	(7, 3) (2, 6)	(-5, 3)	$25+9 = 34$	5.8310	–
P1– P12	(3, 7) (2, 4)	(-1, -3)	$1+9 = 10$	3.1623	–	P5– P11	(7, 3) (3, 5)	(-4, 2)	$16+4 = 20$	4.4721	–
P2– P3	(4, 6) (5, 5)	(1, -1)	$1+1 = 2$	1.4142	✓	P5– P12	(7, 3) (2, 4)	(-5, 1)	$25+1 = 26$	5.0990	–
P2– P4	(4, 6) (6, 4)	(2, -2)	$4+4 = 8$	2.8284	–	P6– P7	(6, 2) (7, 2)	(1, 0)	$1+0 = 1$	1.0000	✓
P2– P5	(4, 6) (7, 3)	(3, -3)	$9+9 = 18$	4.2426	–	P6– P8	(6, 2) (8, 4)	(2, 2)	$4+4 = 8$	2.8284	–
P2– P6	(4, 6) (6, 2)	(2, -4)	$4+16 = 20$	4.4721	–	P6– P9	(6, 2) (9, 5)	(3, 3)	$9+9 = 18$	4.2426	–
P2– P7	(4, 6) (7, 2)	(3, -4)	$9+16 = 25$	5.0000	–	P6– P10	(6, 2) (2, 6)	(-4, 4)	$16+16 = 32$	5.6569	–
P2– P8	(4, 6) (8, 4)	(4, -2)	$16+4 = 20$	4.4721	–	P6– P11	(6, 2) (3, 5)	(-3, 3)	$9+9 = 18$	4.2426	–
P2– P9	(4, 6) (9, 5)	(5, -1)	$25+1 = 26$	5.0990	–	P6– P12	(6, 2) (2, 4)	(-4, 2)	$16+4 = 20$	4.4721	–
P2– P10	(4, 6) (2, 6)	(-2, 0)	$4+0 = 4$	2.0000	–	P7– P8	(7, 2) (8, 4)	(1, 2)	$1+4 = 5$	2.2361	–
P2– P11	(4, 6) (3, 5)	(-1, -1)	$1+1 = 2$	1.4142	✓	P7– P9	(7, 2) (9, 5)	(2, 3)	$4+9 = 13$	3.6056	–
P2– P12	(4, 6) (2, 4)	(-2, -2)	$4+4 = 8$	2.8284	–	P7– P10	(7, 2) (2, 6)	(-5, 4)	$25+16 = 41$	6.4031	–
P3– P4	(5, 5) (6, 4)	(1, -1)	$1+1 = 2$	1.4142	✓	P7– P11	(7, 2) (3, 5)	(-4, 3)	$16+9 = 25$	5.0000	–
P3– P5	(5, 5) (7, 3)	(2, -2)	$4+4 = 8$	2.8284	–	P7– P12	(7, 2) (2, 4)	(-5, 2)	$25+4 = 29$	5.3852	–
P3– P6	(5, 5) (6, 2)	(1, -3)	$1+9 = 10$	3.1623	–	P8– P9	(8, 4) (9, 5)	(1, 1)	$1+1 = 2$	1.4142	✓

PAIR	COORDS	$(\Delta X, \Delta Y)$	$\Delta X^2 + \Delta Y^2$	D	≤ 3.61	PAIR	COORDS	$(\Delta X, \Delta Y)$	$\Delta X^2 + \Delta Y^2$	D	≤ 3.61
P3– P7	(5,5) (7,2)	(2,-3)	4+9 = 13	3.6056	–	P8– P10	(8,4) (2,6)	(-6,2)	36+4 = 40	6.3246	–
P3– P8	(5,5) (8,4)	(3,-1)	9+1 = 10	3.1623	–	P8– P11	(8,4) (3,5)	(-5,1)	25+1 = 26	5.0990	–
P3– P9	(5,5) (9,5)	(4,0)	16+0 = 16	4.0000	–	P8– P12	(8,4) (2,4)	(-6,0)	36+0 = 36	6.0000	–
P3– P10	(5,5) (2,6)	(-3,1)	9+1 = 10	3.1623	–	P9– P10	(9,5) (2,6)	(-7,1)	49+1 = 50	7.0711	–
P3– P11	(5,5) (3,5)	(-2,0)	4+0 = 4	2.0000	–	P9– P11	(9,5) (3,5)	(-6,0)	36+0 = 36	6.0000	–
P3– P12	(5,5) (2,4)	(-3,-1)	9+1 = 10	3.1623	–	P9– P12	(9,5) (2,4)	(-7,-1)	49+1 = 50	7.0711	–
P4– P5	(6,4) (7,3)	(1,-1)	1+1 = 2	1.4142	✓	P10– P11	(2,6) (3,5)	(1,-1)	1+1 = 2	1.4142	✓
P4– P6	(6,4) (6,2)	(0,-2)	0+4 = 4	2.0000	–	P10– P12	(2,6) (2,4)	(0,-2)	0+4 = 4	2.0000	–
P4– P7	(6,4) (7,2)	(1,-2)	1+4 = 5	2.2361	–	P11– P12	(3,5) (2,4)	(-1,-1)	1+1 = 2	1.4142	✓

5 The complete 12×12 distance matrix. It is symmetric with zeros on the diagonal; bold entries are the neighbour distances ≤ 1.9 .

PT	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
P1	0.00	1.41	2.83	4.24	5.66	5.83	6.40	5.83	6.32	1.41	2.00	3.16
P2	1.41	0.00	1.41	2.83	4.24	4.47	5.00	4.47	5.10	2.00	1.41	2.83
P3	2.83	1.41	0.00	1.41	2.83	3.16	3.61	3.16	4.00	3.16	2.00	3.16
P4	4.24	2.83	1.41	0.00	1.41	2.00	2.24	2.00	3.16	4.47	3.16	4.00
P5	5.66	4.24	2.83	1.41	0.00	1.41	1.00	1.41	2.83	5.83	4.47	5.10
P6	5.83	4.47	3.16	2.00	1.41	0.00	1.00	2.83	4.24	5.66	4.24	4.47
P7	6.40	5.00	3.61	2.24	1.00	1.00	0.00	2.24	3.61	6.40	5.00	5.39
P8	5.83	4.47	3.16	2.00	1.41	2.83	2.24	0.00	1.41	6.32	5.10	6.00
P9	6.32	5.10	4.00	3.16	2.83	4.24	3.61	1.41	0.00	7.07	6.00	7.07
P10	1.41	2.00	3.16	4.47	5.83	5.66	6.40	6.32	7.07	0.00	1.41	2.00
P11	2.00	1.41	2.00	3.16	4.47	4.24	5.00	5.10	6.00	1.41	0.00	1.41
P12	3.16	2.83	3.16	4.00	5.10	4.47	5.39	6.00	7.07	2.00	1.41	0.00

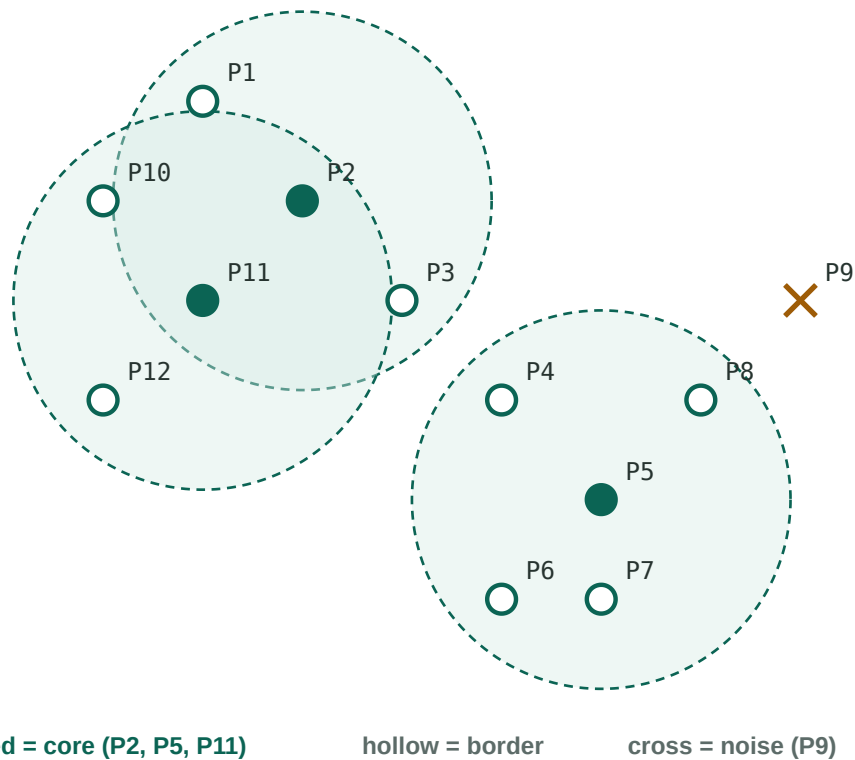
- 6 Read each ε -neighbourhood off the matrix and count it. *MinPts* includes the point itself. Core if the count reaches 4.

POINT	COORD	$N_E(P), D \leq 1.9$	COUNT	$\geq 4?$	CLASS
P1	(3,7)	P1, P2, P10	3	No	Border
P2	(4,6)	P1, P2, P3, P11	4	Yes	Core
P3	(5,5)	P2, P3, P4	3	No	Border
P4	(6,4)	P3, P4, P5	3	No	Border
P5	(7,3)	P4, P5, P6, P7, P8	5	Yes	Core
P6	(6,2)	P5, P6, P7	3	No	Border
P7	(7,2)	P5, P6, P7	3	No	Border
P8	(8,4)	P5, P8, P9	3	No	Border
P9	(9,5)	P8, P9	2	No	Noise
P10	(2,6)	P1, P10, P11	3	No	Border
P11	(3,5)	P2, P10, P11, P12	4	Yes	Core
P12	(2,4)	P11, P12	2	No	Border

P9 is **noise** because it is neither a core point nor inside any core point's neighbourhood — its only neighbour P8 is a border point, not a core point. P12 by contrast has only 2 neighbours but one of them, P11, is a core point, so P12 is a border point.

- 7 Form the clusters. Two core points within ε of each other are density-connected, so their neighbourhoods merge. Check the three cores against each other:

$$d(P_2, P_{11}) = 1.41 \leq 1.9 \Rightarrow \text{merge} \quad d(P_2, P_5) = 4.24 > 1.9 \quad d(P_5, P_{11}) = 4.47 > 1.9$$



The 12 points. Filled markers are core points with their $\varepsilon = 1.9$ neighbourhoods drawn; hollow markers are border points; the cross is noise.

8

Write both conventions — the core-centric one from the video and the formal one.

Perspective A (core-centric): one cluster per core point, border points may repeat.

Cluster 1 (core P2) = {P1, P2, P3, P11}; Cluster 2 (core P5) = {P4, P5, P6, P7, P8}; Cluster 3 (core P11) = {P2, P10, P11, P12}; Noise = {P9}.

Perspective B (formal / standard DBSCAN): merge density-connected cores P2 and P11.

Cluster A = {P1, P2, P3, P10, P11, P12}; Cluster B = {P4, P5, P6, P7, P8}; Noise = {P9}.

Answer Core: P2, P5, P11. Noise: P9. All others are border points. Formally there are **two clusters** — {P1,P2,P3,P10,P11,P12} and {P4,P5,P6,P7,P8} — plus one outlier P9.

Most common error Forgetting that *MinPts* counts the point itself. P2 has neighbours P1, P3, P11 *plus* P2 = 4, which is exactly why it is a core point. Second most common: treating $d = 2.0$ as inside $\epsilon = 1.9$.

Lecture 7A — Chapter 1: Measuring Data Similarity and Dissimilarity

Formulas used in this lecture that have no separate worked problem

NOMINAL ATTRIBUTES

$$D(I, J) = \frac{P - M}{P}$$

M = number of matching attributes, P = total attributes.

Z-SCORE STANDARDISATION

$$Z_{IF} = \frac{X_{IF} - M_F}{S_F}$$

Removes the unit, so kg vs lb no longer changes the answer.

MEAN ABSOLUTE DEVIATION

$$S_F = \frac{1}{N} (|X_{1F} - M_F| + \dots + |X_{NF} - M_F|)$$

Less outlier-sensitive than Σ ; deviations are not squared, so outliers stay detectable.

ORDINAL VARIABLES

$$Z_{IF} = \frac{R_{IF} - 1}{M_F - 1}$$

Rank $R_{IF} \in \{1..M_F\}$ mapped onto $[0, 1]$; then use the interval-scaled methods.

M12 Dissimilarity between asymmetric binary variables

Lecture 7A · 2023 Q7(b)

QUESTION

Calculate the dissimilarity between the asymmetric binary variables in the table below.

NAME	GENDER	FEVER	COUGH	TEST-1	TEST-2	TEST-3	TEST-4
Jack	M	Y	N	P	N	N	N
Mary	F	Y	N	P	N	P	N
Jim	M	Y	P	N	N	N	N

1

Build the contingency table for a pair of objects. q counts attributes where both are 1, r where i is 1 and j is 0, s the reverse, t where both are 0.

	$J = 1$	$J = 0$	SUM
$i = 1$	q	r	$q + r$
$i = 0$	s	t	$s + t$
sum	$q + s$	$r + t$	p

- 2 Three formulas follow from it. For **asymmetric** variables the t cell is **dropped**, because two patients both testing negative tells you nothing.

$$d_{sym}(i, j) = \frac{r + s}{q + r + s + t} \quad d_{asym}(i, j) = \frac{r + s}{q + r + s} \quad sim_{Jaccard}(i, j) = \frac{q}{q + r + s} = 1 - d_{asym}$$

- 3 Encode. **Gender is symmetric, so it is excluded** from an asymmetric calculation. The remaining six attributes are asymmetric binary; set Y and P to 1, N to 0.

over (Fever, Cough, Test-1, Test-2, Test-3, Test-4):

Jack = (1, 0, 1, 0, 0, 0) Mary = (1, 0, 1, 0, 1, 0) Jim = (1, 1, 0, 0, 0, 0)

- 4 **Jack vs Mary.** Both 1 on Fever and Test-1 $\rightarrow q = 2$. Jack 1 / Mary 0: none $\rightarrow r = 0$. Jack 0 / Mary 1: Test-3 $\rightarrow s = 1$.

$$d(\text{jack}, \text{mary}) = \frac{0 + 1}{2 + 0 + 1} = \frac{1}{3} = 0.33$$

- 5 **Jack vs Jim.** Both 1 on Fever $\rightarrow q = 1$. Jack 1 / Jim 0: Test-1 $\rightarrow r = 1$. Jack 0 / Jim 1: Cough $\rightarrow s = 1$.

$$d(\text{jack}, \text{jim}) = \frac{1 + 1}{1 + 1 + 1} = \frac{2}{3} = 0.67$$

- 6 **Jim vs Mary.** Both 1 on Fever $\rightarrow q = 1$. Jim 1 / Mary 0: Cough $\rightarrow r = 1$. Jim 0 / Mary 1: Test-1 and Test-3 $\rightarrow s = 2$.

$$d(\text{jim}, \text{mary}) = \frac{1 + 2}{1 + 1 + 2} = \frac{3}{4} = 0.75$$

Answer $d(\text{jack}, \text{mary}) = 0.33$, $d(\text{jack}, \text{jim}) = 0.67$, $d(\text{jim}, \text{mary}) = 0.75$. Jack and Mary are most similar and most likely to have the same disease; Jim and Mary are least alike.

Two traps (1) Including Gender — it is symmetric and must be removed first. (2) Including t in the denominator — for asymmetric variables the 0–0 agreements are never counted.

M13 Minkowski distance — L_1 and L_2 dissimilarity matrices

Lecture 7A

QUESTION

Define the Minkowski distance and its metric properties, then compute the Manhattan (L_1) and Euclidean (L_2) dissimilarity matrices for the four points $x_1(1, 2)$, $x_2(3, 5)$, $x_3(2, 0)$, $x_4(4, 5)$.

- 1 General form, for p -dimensional objects and order h :

$$d(i, j) = \sqrt[h]{|x_{i1} - x_{j1}|^h + |x_{i2} - x_{j2}|^h + \cdots + |x_{ip} - x_{jp}|^h}$$

A distance satisfying positive definiteness ($d(i, j) > 0$ if $i \neq j$, $d(i, i) = 0$), symmetry ($d(i, j) = d(j, i)$) and the triangle inequality ($d(i, j) \leq d(i, k) + d(k, j)$) is a **metric**.

- 2 $h = 1$ gives **Manhattan** (city-block); $h = 2$ gives **Euclidean**.

$$d_{L1}(i, j) = |x_{i1} - x_{j1}| + \cdots + |x_{ip} - x_{jp}| \quad d_{L2}(i, j) = \sqrt{|x_{i1} - x_{j1}|^2 + \cdots + |x_{ip} - x_{jp}|^2}$$

- 3 Compute all six pairs. Take the coordinate differences once and reuse them for both norms.

$$\begin{aligned}
 x_2, x_1 : \Delta = (2, 3) &\Rightarrow L_1 = 2 + 3 = 5, & L_2 = \sqrt{4 + 9} = \sqrt{13} = 3.61 \\
 x_3, x_1 : \Delta = (1, 2) &\Rightarrow L_1 = 1 + 2 = 3, & L_2 = \sqrt{1 + 4} = \sqrt{5} = 2.24 \\
 x_3, x_2 : \Delta = (1, 5) &\Rightarrow L_1 = 1 + 5 = 6, & L_2 = \sqrt{1 + 25} = \sqrt{26} = 5.10 \\
 x_4, x_1 : \Delta = (3, 3) &\Rightarrow L_1 = 3 + 3 = 6, & L_2 = \sqrt{9 + 9} = \sqrt{18} = 4.24 \\
 x_4, x_2 : \Delta = (1, 0) &\Rightarrow L_1 = 1 + 0 = 1, & L_2 = \sqrt{1 + 0} = 1.00 \\
 x_4, x_3 : \Delta = (2, 5) &\Rightarrow L_1 = 2 + 5 = 7, & L_2 = \sqrt{4 + 25} = \sqrt{29} = 5.39
 \end{aligned}$$

- 4 Assemble the two lower-triangular matrices (the diagonal is 0 and the matrix is symmetric, so only the lower half is written).

L_1	X_1	X_2	X_3	X_4
x_1	0			
x_2	5	0		
x_3	3	6	0	
x_4	6	1	7	0

L_2	X_1	X_2	X_3	X_4
x_1	0			
x_2	3.61	0		
x_3	2.24	5.10	0	
x_4	4.24	1.00	5.39	0

Answer The two matrices above. Closest pair x_2, x_4 ; farthest pair x_3, x_4 .

Checks $d_{L2} \leq d_{L1}$ always — the Euclidean value can never exceed the Manhattan one. And $h \rightarrow \infty$ gives the supremum distance $\max_f |x_{if} - x_{jf}|$; quote it for a bonus mark.

M14 Cosine similarity between two documents

Lecture 7A

QUESTION

Find the similarity between documents d_1 and d_2 , given as term-frequency vectors:

$$d_1 = (5, 0, 3, 0, 2, 0, 0, 2, 0, 0) \quad d_2 = (3, 0, 2, 0, 1, 1, 0, 1, 0, 1)$$

- 1 Cosine similarity is the dot product normalised by both vector lengths.

$$\cos(d_1, d_2) = \frac{d_1 \bullet d_2}{\|d_1\| \|d_2\|}$$

- 2 Dot product — multiply position by position and add. Only positions where *both* are non-zero contribute.

$$\begin{aligned}
 d_1 \bullet d_2 &= 5 \cdot 3 + 0 \cdot 0 + 3 \cdot 2 + 0 \cdot 0 + 2 \cdot 1 + 0 \cdot 1 + 0 \cdot 0 + 2 \cdot 1 + 0 \cdot 0 + 0 \cdot 1 \\
 &= 15 + 6 + 2 + 2 = 25
 \end{aligned}$$

- 3 Length of d_1 — root of the sum of squares.

$$\|d_1\| = (5^2 + 3^2 + 2^2 + 2^2)^{0.5} = (25 + 9 + 4 + 4)^{0.5} = (42)^{0.5} = 6.481$$

- 4 Length of d_2 .

$$\|d_2\| = (3^2 + 2^2 + 1^2 + 1^2 + 1^2 + 1^2)^{0.5} = (9 + 4 + 1 + 1 + 1 + 1)^{0.5} = (17)^{0.5} = 4.12$$

- 5 Divide.

$$\cos(d_1, d_2) = \frac{25}{6.481 \times 4.12} = \frac{25}{26.70} = 0.9356 \approx 0.94$$

Answer $\cos(d_1, d_2) = 0.94$ — the two documents are highly similar in orientation.

Why cosine It ignores magnitude and measures only direction, so a long and a short document on the same topic still score as similar. $\cos = 1$ is identical direction, $\cos = 0$ is no shared terms.

Lecture 7B — Chapter 4: Mining Frequent Patterns, Associations and Correlations

Problems M18, M20, M22 and M23 share one 9-transaction database; M16 and M19 share the 4-transaction A–E database. Each problem prints the database it uses, so every problem is self-contained.

M15 Support, confidence and strong rules

Lecture 7B

QUESTION

For the transaction database below with $\text{minsup} = 50\%$ and $\text{minconf} = 50\%$, list the frequent patterns and two association rules with their support and confidence.

TID	ITEMS BOUGHT
10	Beer, Nuts, Diaper
20	Beer, Coffee, Diaper
30	Beer, Diaper, Eggs
40	Nuts, Eggs, Milk
50	Nuts, Coffee, Diaper, Eggs, Milk

- Definitions. **Absolute support** (support count) of X is its occurrence count; **relative support** s is the fraction of transactions containing X . X is **frequent** when its support reaches minsup .
- For the rule $X \Rightarrow Y$:

$$\text{support}(X \Rightarrow Y) = P(X \cup Y) \quad \text{confidence}(X \Rightarrow Y) = P(Y | X) = \frac{\text{support_count}(X \cup Y)}{\text{support_count}(X)}$$
- Convert the threshold to a count: 50% of 5 transactions is 2.5, so a frequent itemset needs a count ≥ 3 . Count each item:

Beer 3, Nuts 3, Diaper 4, Eggs 3, Milk 2, Coffee 2

Milk and Coffee fall below 3. Among pairs, only $\{\text{Beer}, \text{Diaper}\}$ reaches 3 (Tid 10, 20, 30).

Frequent patterns: Beer:3, Nuts:3, Diaper:4, Eggs:3, $\{\text{Beer}, \text{Diaper}\}$:3
- Generate two rules from $\{\text{Beer}, \text{Diaper}\}$. Support is the same for both; confidence differs because the denominator differs.

$$\text{Beer} \Rightarrow \text{Diaper} : \text{support} = \frac{3}{5} = 60\%, \quad \text{conf} = \frac{\text{support_count}(\{\text{Beer}, \text{Diaper}\})}{\text{support_count}(\text{Beer})} = \frac{3}{3} = 100\%$$

$$\text{Diaper} \Rightarrow \text{Beer} : \text{support} = \frac{3}{5} = 60\%, \quad \text{conf} = \frac{3}{4} = 75\%$$
- A rule satisfying **both** minsup and minconf is a **strong rule**. Both rules clear 50% support and 50% confidence, so both are strong.

Answer Frequent: Beer:3, Nuts:3, Diaper:4, Eggs:3, {Beer,Diaper}:3. Rules: Beer $\Rightarrow$ Diaper (60%, 100%) and Diaper $\Rightarrow$ Beer (60%, 75%) — both strong.

Remember Support is **symmetric** (both directions share it); confidence is **not**. That asymmetry is exactly why both numbers must be quoted.

M16 Apriori algorithm — A, B, C, D, E database

Lecture 7B

QUESTION

Using the Apriori algorithm find all frequent itemsets in the database below with $Sup_{min} = 2$.

TID	ITEMS
T10	A, C, D
T20	B, C, E
T30	A, B, C, E
T40	B, E

- 1 Apriori property (downward closure).** Any subset of a frequent itemset must also be frequent — equivalently, if an itemset is infrequent, none of its supersets can be frequent, so they are never generated or tested.
- 2 Scan 1 — $C_1 \rightarrow L_1$.** Count each single item and prune below 2.

C_1	TRANSACTIONS	COUNT	STATUS
{A}	T10, T30	2	Keep
{B}	T20, T30, T40	3	Keep
{C}	T10, T20, T30	3	Keep
{D}	T10	1	Pruned
{E}	T20, T30, T40	3	Keep

$$L_1 = \{ \{A\}:2, \{B\}:3, \{C\}:3, \{E\}:3 \}$$

- 3 Scan 2 — C_2 by self-join $L_1 \bowtie L_1$.** All pairs of surviving items; note D never appears again.

$$C_2 = \{ \{A, B\}, \{A, C\}, \{A, E\}, \{B, C\}, \{B, E\}, \{C, E\} \}$$

C_2	FOUND IN	COUNT	STATUS
{A, B}	T30	1	Pruned
{A, C}	T10, T30	2	Keep
{A, E}	T30	1	Pruned
{B, C}	T20, T30	2	Keep
{B, E}	T20, T30, T40	3	Keep
{C, E}	T20, T30	2	Keep

$$L_2 = \{ \{A, C\}:2, \{B, C\}:2, \{B, E\}:3, \{C, E\}:2 \}$$

- 4 **Scan 3** — C_3 : **join, then prune, then count.** Join itemsets sharing their first $k - 1$ items.

$$\{B, C\} \bowtie \{B, E\} \rightarrow \{B, C, E\} \quad \{A, C\} \bowtie \{B, C\} \rightarrow \{A, B, C\}$$

Prune step — every $(k-1)$ -subset must be in L_2 :

For $\{A, B, C\}$: subsets $\{A, B\}, \{A, C\}, \{B, C\}$. Since $\{A, B\} \notin L_2$, $\{A, B, C\}$ is **pruned without any database scan**.

For $\{B, C, E\}$: subsets $\{B, C\}, \{B, E\}, \{C, E\}$ — all in L_2 , so it is retained.

$$C_3 = \{\{B, C, E\}\} \rightarrow \text{found in T20, T30, count } 2 \geq 2 \\ L_3 = \{\{B, C, E\}:2\}$$

- 5 **Termination.** L_3 has only one itemset, and at least two 3-itemsets are needed to form a 4-itemset, so C_4 cannot be built. The algorithm stops.

Answer 1-itemsets $\{A\}, \{B\}, \{C\}, \{E\}$; 2-itemsets $\{A, C\}, \{B, C\}, \{B, E\}, \{C, E\}$; 3-itemset $\{B, C, E\}$.

Show the prune step separately Marks are given for demonstrating that $\{A, B, C\}$ died on the subset test *before* any scan — that is the whole point of the Apriori property. A second standard example: $L_3 = \{abc, abd, acd, ace, bcd\}$ joins to $abcd$ and $acde$; $acde$ is pruned because $ade \notin L_3$, so $C_4 = \{abcd\}$.

M17 Apriori algorithm — Beer / Nuts / Diaper database

Lecture 7B · 2023 Q4(c)

QUESTION

Using the Apriori algorithm and the dataset below, find the frequent itemsets where $Min\text{support} = 2$.

TID	ITEMS BOUGHT
10	Beer, Nuts, Diaper
20	Beer, Coffee, Diaper
30	Beer, Diaper, Eggs
40	Nuts, Eggs, Milk
50	Nuts, Coffee, Diaper, Eggs, Milk

Same database as M15, but the threshold is now a **count of 2** rather than 50%, so more itemsets survive.

- 1 **Scan 1** — $C_1 \rightarrow L_1$. With $minsup = 2$, every one of the six items survives.

ITEM	TRANSACTIONS	COUNT	STATUS
Beer	10, 20, 30	3	Keep
Nuts	10, 40, 50	3	Keep
Diaper	10, 20, 30, 50	4	Keep
Coffee	20, 50	2	Keep
Eggs	30, 40, 50	3	Keep
Milk	40, 50	2	Keep

2 Scan 2 — C_2 . Six items give $\binom{6}{2} = 15$ candidate pairs. Count each.

PAIR	FOUND IN	COUNT	STATUS
{Beer, Coffee}	20	1	Pruned
{Beer, Diaper}	10, 20, 30	3	Keep
{Beer, Eggs}	30	1	Pruned
{Beer, Milk}	—	0	Pruned
{Beer, Nuts}	10	1	Pruned
{Coffee, Diaper}	20, 50	2	Keep
{Coffee, Eggs}	50	1	Pruned
{Coffee, Milk}	50	1	Pruned
{Coffee, Nuts}	50	1	Pruned
{Diaper, Eggs}	30, 50	2	Keep
{Diaper, Milk}	50	1	Pruned
{Diaper, Nuts}	10, 50	2	Keep
{Eggs, Milk}	40, 50	2	Keep
{Eggs, Nuts}	40, 50	2	Keep
{Milk, Nuts}	40, 50	2	Keep

$$L_2 = \{\{Beer, Diaper\}:3, \{Coffee, Diaper\}:2, \{Diaper, Eggs\}:2, \{Diaper, Nuts\}:2, \{Eggs, Milk\}:2, \{Eggs, Nuts\}:2, \{Milk, Nuts\}:2\}$$

3 C_3 — join then prune. Only two triples have all three of their pairs in L_2 :

$\{Diaper, Eggs, Nuts\}$ — subsets $\{Diaper, Eggs\}$, $\{Diaper, Nuts\}$, $\{Eggs, Nuts\}$ all present → candidate.

$\{Eggs, Milk, Nuts\}$ — subsets $\{Eggs, Milk\}$, $\{Eggs, Nuts\}$, $\{Milk, Nuts\}$ all present → candidate.

Everything containing Beer or Coffee dies at the subset test, e.g. $\{Beer, Diaper, Nuts\}$ needs $\{Beer, Nuts\} \notin L_2$.

4 Scan 3.

CANDIDATE	FOUND IN	COUNT	STATUS
{Diaper, Eggs, Nuts}	50	1	Pruned
{Eggs, Milk, Nuts}	40, 50	2	Keep

$$L_3 = \{\{Eggs, Milk, Nuts\}:2\}$$

L_3 has one itemset, so no C_4 can be formed. Terminate.

Answer L_1 = all six items; L_2 = the seven pairs listed above; L_3 = $\{Eggs, Milk, Nuts\}:2$, which is the maximal frequent itemset.

QUESTION

Given the frequent itemset $L = \{I_1, I_2, I_5\}$ from the 9-transaction database below, generate all association rules and identify the strong ones with $\text{minconf} = 70\%$.

TID	LIST OF ITEM IDS	TID	LIST OF ITEM IDS	TID	LIST OF ITEM IDS
T100	I1, I2, I5	T400	I1, I2, I4	T700	I1, I3
T200	I2, I4	T500	I1, I3	T800	I1, I2, I3, I5
T300	I2, I3	T600	I2, I3	T900	I1, I2, I3

- 1 Procedure: for the frequent itemset L , take every non-empty **proper** subset $S \subset L$ and output the rule $S \Rightarrow (L - S)$, keeping it if its confidence reaches the threshold.

$$\text{Confidence}(S \Rightarrow (L - S)) = \frac{\text{support_count}(L)}{\text{support_count}(S)} \geq \text{min_conf}$$

A k -itemset gives exactly $2^k - 2$ rules; here $2^3 - 2 = 6$.

- 2 Read all the required support counts off the database once. **The numerator is the same for every rule** — only the denominator changes.

$$\begin{aligned} \text{support_count}(\{I_1, I_2, I_5\}) &= 2 \quad (\text{T100, T800}) \\ \text{support_count}(\{I_1, I_2\}) &= 4 \quad (\text{T100, T400, T800, T900}) \\ \text{support_count}(\{I_1, I_5\}) &= 2 \quad (\text{T100, T800}) \quad \text{support_count}(\{I_2, I_5\}) = 2 \quad (\text{T100, T800}) \\ \text{support_count}(\{I_1\}) &= 6 \quad \text{support_count}(\{I_2\}) = 7 \quad \text{support_count}(\{I_5\}) = 2 \end{aligned}$$

- 3 Divide for each of the six subsets.

$$\begin{aligned} \text{R1} \quad I_1 \wedge I_2 &\Rightarrow I_5 : \quad \text{conf} = \frac{2}{4} = 50\% \\ \text{R2} \quad I_1 \wedge I_5 &\Rightarrow I_2 : \quad \text{conf} = \frac{2}{2} = 100\% \\ \text{R3} \quad I_2 \wedge I_5 &\Rightarrow I_1 : \quad \text{conf} = \frac{2}{2} = 100\% \\ \text{R4} \quad I_1 &\Rightarrow I_2 \wedge I_5 : \quad \text{conf} = \frac{2}{6} = 33.33\% \approx 33\% \\ \text{R5} \quad I_2 &\Rightarrow I_1 \wedge I_5 : \quad \text{conf} = \frac{2}{7} = 28.57\% \approx 29\% \\ \text{R6} \quad I_5 &\Rightarrow I_1 \wedge I_2 : \quad \text{conf} = \frac{2}{2} = 100\% \end{aligned}$$

- 4 Keep those at or above 70%.

Answer Strong rules: $I_1 \wedge I_5 \Rightarrow I_2$ (100%), $I_2 \wedge I_5 \Rightarrow I_1$ (100%), $I_5 \Rightarrow I_1 \wedge I_2$ (100%). R1, R4 and R5 fail the 70% threshold.

QUESTION

For the database below, find the frequent patterns using the hash-based technique (DHP). Minimum support is 2.

TID	ITEMS
T10	A, C, D
T20	B, C, E
T30	A, B, C, E
T40	B, E

- Idea.** During the first scan, as well as counting single items, hash every 2-item pair of each transaction into a bucket and increment that bucket's counter. If a bucket's total is below *minsup*, **no** pair inside it can be frequent, so all of them are pruned from C_2 before the second scan begins.
- Scan 1 — $C_1 \rightarrow L_1$, with the pruning table.** Count every single item across the four transactions and prune anything below 2.

C_1 ITEMSET	TRANSACTIONS CONTAINING IT	SUPPORT COUNT	STATUS (≥ 2)
{A}	T10, T30	2	Frequent – keep
{B}	T20, T30, T40	3	Frequent – keep
{C}	T10, T20, T30	3	Frequent – keep
{D}	T10	1	Infrequent – pruned
{E}	T20, T30, T40	3	Frequent – keep

$$L_1 = \{A, B, C, E\}$$

D is dropped here and never reappears — but note that pairs *containing* D were still hashed during this same scan, which is why bucket counts below can include $\{A, D\}$ and $\{C, D\}$.

- The hash function.** Give each item an integer order $A=1, B=2, C=3, D=4, E=5$. For a pair $\{X, Y\}$ written in order,

$$\text{Hash}(X, Y) = (\text{order}(X) \times 10 + \text{order}(Y)) \bmod 7$$

- Extract every pair from every transaction and hash it.** A transaction of k items gives $\binom{k}{2}$ pairs.

T10 {A,C,D} → 3 pairs

$$\{A, C\} \rightarrow 1 \times 10 + 3 = 13 \bmod 7 = 6 \quad \{A, D\} \rightarrow 14 \bmod 7 = 0 \quad \{C, D\} \rightarrow 34 \bmod 7 = 6$$

T20 {B,C,E} → 3 pairs

$$\{B, C\} \rightarrow 23 \bmod 7 = 2 \quad \{B, E\} \rightarrow 25 \bmod 7 = 4 \quad \{C, E\} \rightarrow 35 \bmod 7 = 0$$

T30 {A,B,C,E} → 6 pairs

$$\{A, B\} \rightarrow 12 \bmod 7 = 5 \quad \{A, C\} \rightarrow 6 \quad \{A, E\} \rightarrow 15 \bmod 7 = 1 \quad \{B, C\} \rightarrow 2 \quad \{B, E\} \rightarrow 4 \quad \{C, E\} \rightarrow$$

T40 {B,E} → 1 pair: $\{B, E\} \rightarrow 4$. Total pairs hashed: $3 + 3 + 6 + 1 = 13$.

- 5 **Bucket table and bit vector.** A bucket qualifies when its *total* count reaches 2.

BUCKET	ITEMSETS HASHED INTO IT	BUCKET COUNT	BIT (COUNT ≥ 2)
0	{C,E} T20, {C,E} T30, {A,D} T10	3	1
1	{A,E} T30	1	0
2	{B,C} T20, {B,C} T30	2	1
3	(empty)	0	0
4	{B,E} T20, {B,E} T30, {B,E} T40	3	1
5	{A,B} T30	1	0
6	{A,C} T10, {C,D} T10, {A,C} T30	3	1

binary vector = [1, 0, 1, 0, 1, 0, 1]

- 6 **Prune C_2 .** Standard Apriori would test 6 candidates; check each against its bucket's bit.

CANDIDATE	BUCKET	BIT	ACTION
{A,B}	5 (count 1)	0	Pruned
{A,C}	6 (count 3)	1	Kept
{A,E}	1 (count 1)	0	Pruned
{B,C}	2 (count 2)	1	Kept
{B,E}	4 (count 3)	1	Kept
{C,E}	0 (count 3)	1	Kept

$$C_2 = \{\{A,C\}, \{B,C\}, \{B,E\}, \{C,E\}\}$$

- 7 Continue with standard Apriori. Scanning these four gives $L_2 = \{\{A,C\}:2, \{B,C\}:2, \{B,E\}:3, \{C,E\}:2\}$ and then $L_3 = \{\{B,C,E\}:2\}$, exactly as in M16.

Answer DHP reduces $|C_2|$ from 6 to 4 before the second scan; the frequent patterns are {A}, {B}, {C}, {E}; {A,C}, {B,C}, {B,E}, {C,E}; {B,C,E}.

A bucket count is a combined count Bucket 0 reaches 3 only because {C,E} (twice) and {A,D} (once) collided. A bucket passing the threshold does **not** prove any pair inside it is frequent — DHP is a pruning filter, never a support oracle. Survivors still need a real scan.

M20 DHP on the 9-transaction I1–I5 database

Lecture 7B

QUESTION

Apply the hash-based technique (DHP) to the transaction database below with **minimum support count 3**. Item order is $I_1=1, I_2=2, I_3=3, I_4=4, I_5=5$ and the hash function is $H(x,y) = (\text{order}(x) \times 10 + \text{order}(y)) \bmod 7$.

TID	LIST OF ITEMS	TID	LIST OF ITEMS	TID	LIST OF ITEMS
T1	I1, I2, I5	T4	I1, I2, I4	T7	I1, I3
T2	I2, I4	T5	I1, I3	T8	I1, I2, I3, I5
T3	I2, I3	T6	I2, I3	T9	I1, I2, I3

- 1 **Scan 1** — $C_1 \rightarrow L_1$. Count each item and apply the threshold of 3.

ITEM	TRANSACTIONS CONTAINING IT	SUPPORT COUNT	STATUS (≥ 3)
I1	T1, T4, T5, T7, T8, T9	6	Frequent – keep
I2	T1, T2, T3, T4, T6, T8, T9	7	Frequent – keep
I3	T3, T5, T6, T7, T8, T9	6	Frequent – keep
I4	T2, T4	2	Infrequent – pruned
I5	T1, T8	2	Infrequent – pruned

$$L_1 = \{I_1, I_2, I_3\}$$

Note the contrast with M22/M23, where $min_sup = 2$ keeps all five items. Raising the threshold to 3 removes I_4 and I_5 .

- 2 **Hash every pair.** Count each pair across the database, then compute its bucket with $H(x, y)$.

PAIR	OCCURRING IN	COUNT	HASH CALCULATION	BUCKET
{I1, I2}	T1, T4, T8, T9	4	$(1 \times 10 + 2) = 12 \bmod 7$	5
{I1, I3}	T5, T7, T8, T9	4	$(1 \times 10 + 3) = 13 \bmod 7$	6
{I1, I4}	T4	1	$(1 \times 10 + 4) = 14 \bmod 7$	0
{I1, I5}	T1, T8	2	$(1 \times 10 + 5) = 15 \bmod 7$	1
{I2, I3}	T3, T6, T8, T9	4	$(2 \times 10 + 3) = 23 \bmod 7$	2
{I2, I4}	T2, T4	2	$(2 \times 10 + 4) = 24 \bmod 7$	3
{I2, I5}	T1, T8	2	$(2 \times 10 + 5) = 25 \bmod 7$	4
{I3, I4}	(none)	0	–	–
{I3, I5}	T8	1	$(3 \times 10 + 5) = 35 \bmod 7$	0
{I4, I5}	(none)	0	–	–

- 3 **Bucket table.** Sum the counts landing in each bucket; qualify when the total reaches 3.

BUCKET	0	1	2	3	4	5	6
Contents	{I1, I4}:1, {I3, I5}:1	{I1, I5}:2	{I2, I3}:4	{I2, I4}:2	{I2, I5}:2	{I1, I2}:4	{I1, I3}:4
Total	1+1 = 2	2	4	2	2	4	4
Qualifies (≥ 3)	No	No	Yes	No	No	Yes	Yes

- 4 **Candidate filtering.** Joining L_1 gives three candidates; check each against its bucket.

$$\{I_1, I_2\} \rightarrow \text{bucket 5, } 4 \geq 3 \checkmark \quad \{I_1, I_3\} \rightarrow \text{bucket 6, } 4 \geq 3 \checkmark \quad \{I_2, I_3\} \rightarrow \text{bucket 2, } 4 \geq 3 \checkmark$$

$$L_2 = \{ \{I_1, I_2\}, \{I_1, I_3\}, \{I_2, I_3\} \}$$

Answer $L_1 = \{I_1, I_2, I_3\}$ and $L_2 = \{ \{I_1, I_2\}, \{I_1, I_3\}, \{I_2, I_3\} \}$, each with count 4.

DHP advantages Reduces the number of scans and removes the large candidates that cause high I/O cost.

QUESTION

Explain transaction reduction and apply it to the database below with $min_sup = 2$.

TRANS.	ITEMS
T1	I1, I2, I5
T2	I2, I3, I4
T3	I3, I4
T4	I1, I2, I3, I4

- 1 **Core rule.** A transaction that does not contain any frequent k -itemset cannot contain any frequent $(k+1)$ -itemset, so it can be marked or removed from later scans.

- 2 L_1 . Count each item across the 4 transactions.

ITEM	PRESENT IN	COUNT	STATUS (≥ 2)
I1	T1, T4	2	Keep
I2	T1, T2, T4	3	Keep
I3	T2, T3, T4	3	Keep
I4	T2, T3, T4	3	Keep
I5	T1	1	Removed

After deleting I_5 : T1: {I1,I2}, T2: {I2,I3,I4}, T3: {I3,I4}, T4: {I1,I2,I3,I4}.

- 3 L_2 . Six candidate pairs from $\{I_1, I_2, I_3, I_4\}$.

PAIR	PRESENT IN	COUNT	STATUS
{I1, I2}	T1, T4	2	Frequent
{I1, I3}	T4	1	Pruned
{I1, I4}	T4	1	Pruned
{I2, I3}	T2, T4	2	Frequent
{I2, I4}	T2, T4	2	Frequent
{I3, I4}	T2, T3, T4	3	Frequent

$$L_2 = \{\{I_1, I_2\}, \{I_2, I_3\}, \{I_2, I_4\}, \{I_3, I_4\}\}$$

- 4 **Reduce the database for $k = 3$.** A transaction must retain at least 3 frequent items to contain any 3-itemset.

T1 retains only $\{I_1, I_2\}$ (length 2) → **removed**.

T3 retains only $\{I_3, I_4\}$ (length 2) → **removed**.

T2 retains $\{I_2, I_3, I_4\}$ (length 3) → retained.

T4 retains $\{I_1, I_2, I_3, I_4\}$ (length 4) → retained.

The database shrinks from 4 transactions to 2 for the third pass.

- 5 L_3 . The only candidate from L_2 is $\{I_2, I_3, I_4\}$, tested against T2 and T4 only.

$$T2 \supseteq \{I_2, I_3, I_4\} \rightarrow 1 \quad T4 \supseteq \{I_2, I_3, I_4\} \rightarrow 1 \quad \text{count} = 2 \geq 2$$

$$L_3 = \{\{I_2, I_3, I_4\}\}$$

Answer $L_3 = \{I_2, I_3, I_4\}$ with support 2, found by scanning only 2 of the 4 transactions on the third pass.

QUESTION

Construct the FP-tree for the transaction database below with $min_sup = 2$ and mine all frequent patterns.

TID	ITEMS	TID	ITEMS	TID	ITEMS
T100	I1, I2, I5	T400	I1, I2, I4	T700	I1, I3
T200	I2, I4	T500	I1, I3	T800	I1, I2, I3, I5
T300	I2, I3	T600	I2, I3	T900	I1, I2, I3

- 1 **Scan 1 — count items, sort descending.** This sorted list is the **header table**, and it fixes the order used everywhere afterwards.

ITEM	TRANSACTIONS CONTAINING IT	COUNT	RANK
I2	T100, T200, T300, T400, T600, T800, T900	7	1st
I1	T100, T400, T500, T700, T800, T900	6	2nd
I3	T300, T500, T600, T700, T800, T900	6	3rd
I4	T200, T400	2	4th
I5	T100, T800	2	5th

order: $I_2 \rightarrow I_1 \rightarrow I_3 \rightarrow I_4 \rightarrow I_5$

Sorting descending is what makes transactions share branches — the most common item becomes the trunk. Nothing is pruned here: I_4 and I_5 sit exactly at the threshold.

- 2 **Re-order every transaction by that rank.** Do this on paper before drawing anything.

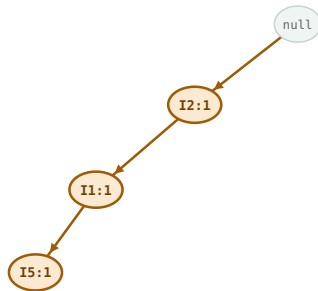
TID	ORIGINAL	RE-ORDERED	TID	ORIGINAL	RE-ORDERED
T100	I1, I2, I5	I2, I1, I5	T600	I2, I3	I2, I3
T200	I2, I4	I2, I4	T700	I1, I3	I1, I3
T300	I2, I3	I2, I3	T800	I1, I2, I3, I5	I2, I1, I3, I5
T400	I1, I2, I4	I2, I1, I4	T900	I1, I2, I3	I2, I1, I3
T500	I1, I3	I1, I3			

3

Insert each transaction, one at a time, from the null root. If the prefix already exists, walk it and add 1 to every node on it; otherwise branch out with new nodes at count 1. The nine drawings below are the same tree at nine moments — nodes never move, they only appear or their counter changes.

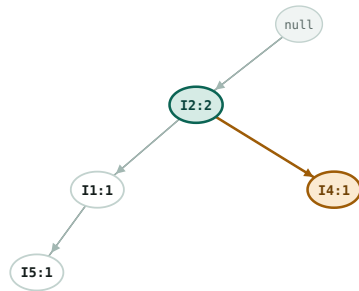
 node created (count set to 1)
 count incremented
 untouched this step

1 · T100 I2, I1, I5



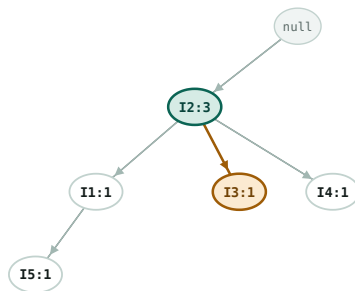
Empty tree — nothing to share, so all three nodes are created at count 1.

2 · T200 I2, I4



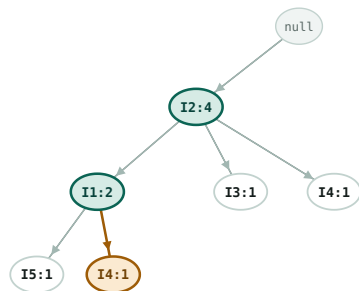
Prefix $\langle I_2 \rangle$ already exists: $1 \rightarrow 2$. I_4 is new — I_2 's second child.

3 · T300 I2, I3



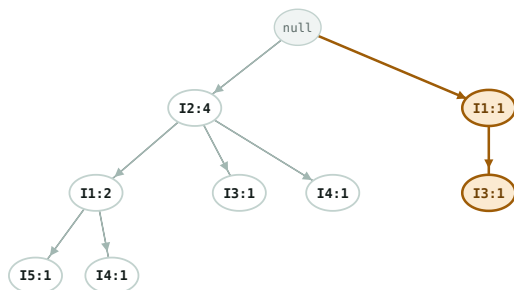
Again $\langle I_2 \rangle$: $2 \rightarrow 3$. I_3 becomes a third child of I_2 .

4 · T400 I2, I1, I4



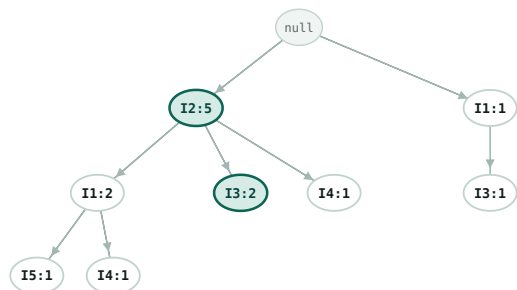
Prefix $\langle I_2, I_1 \rangle$ matches: $3 \rightarrow 4$ and $1 \rightarrow 2$. A **second** I_4 node appears — same item, different path.

5 · T500 I1, I3



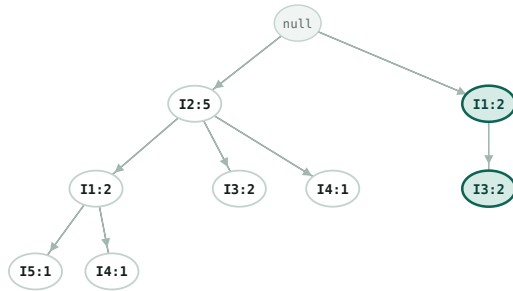
No I_2 , so the matching prefix is empty — start a **new branch off the root**. This is the step students lose marks on.

6 · T600 I2, I3



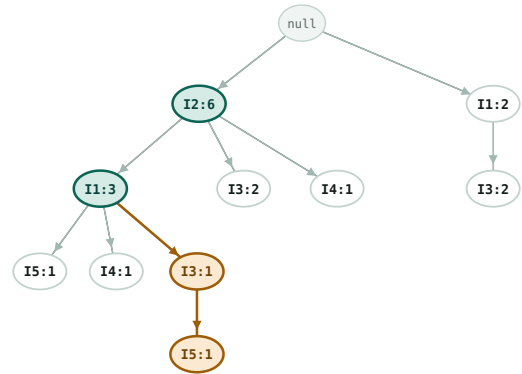
The whole transaction matches an existing path: $4 \rightarrow 5$, $1 \rightarrow 2$. **No new nodes at all.**

7 · T700 I1, I3



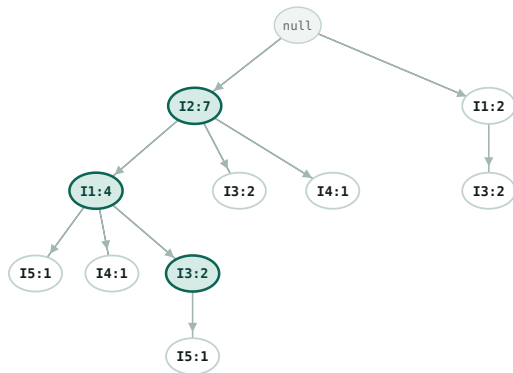
Same again on the right-hand branch: $1 \rightarrow 2$ and $1 \rightarrow 2$.

8 · T800 I2, I1, I3, I5



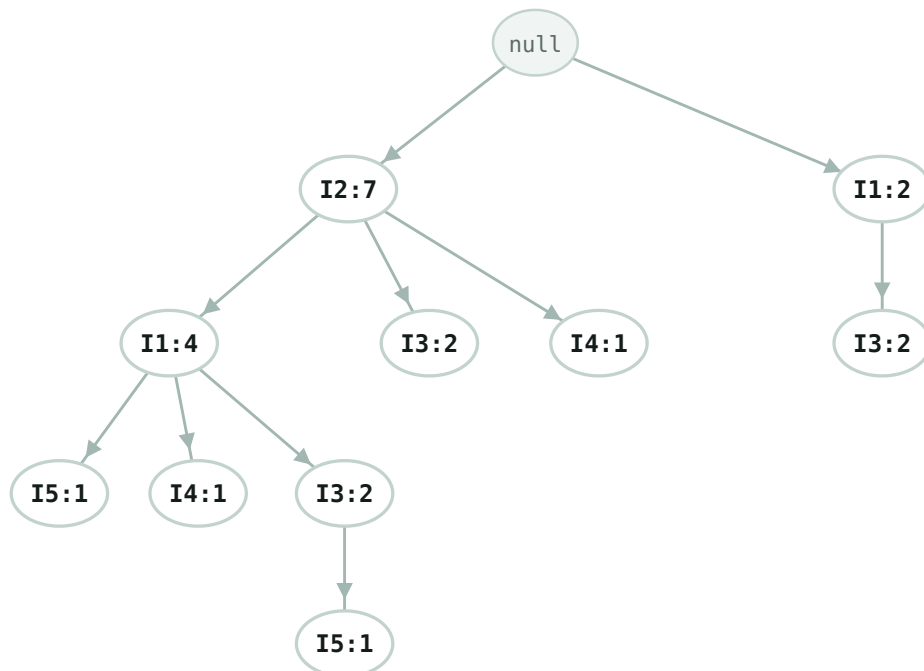
Prefix $\langle I_2, I_1 \rangle$ matches; then I_3 and I_5 are created. Second I_5 node, deeper in the tree.

9 · T900 I2, I1, I3



Every node already exists: $6 \rightarrow 7$, $3 \rightarrow 4$, $1 \rightarrow 2$. **Tree complete.**

4 The finished FP-tree. Nine transactions are now stored in ten nodes, and the database is never read again.



Final FP-tree with the header-table order $I_2 \rightarrow I_1 \rightarrow I_3 \rightarrow I_4 \rightarrow I_5$.

- 5 **Check before mining.** Sum the counts of every node carrying the same item; each total must equal that item's support from Step 1.

$$I_2 : 7 = 7 \checkmark \quad I_1 : 4 + 2 = 6 \checkmark \quad I_3 : 2 + 2 + 2 = 6 \checkmark \quad I_4 : 1 + 1 = 2 \checkmark \quad I_5 : 1 + 1 = 2 \checkmark$$

- 6 **Mine bottom-up** — least frequent suffix first: $I_5 \rightarrow I_4 \rightarrow I_3 \rightarrow I_1$. For each suffix ask three questions: (i) what prefix paths lie above its nodes (the **conditional pattern base**)? (ii) summing item counts over those paths, which items reach min_sup (the **conditional FP-tree**)? (iii) what patterns come from combining survivors with the suffix?

Each prefix path carries the count of the suffix node, never the larger count written on an ancestor.

- 7 **Suffix I_5 .** Two nodes.

Path 1: $null \rightarrow (I_2:7) \rightarrow (I_1:4) \rightarrow (I_5:1) \Rightarrow$ prefix $\{I_2, I_1\}$ count 1

Path 2: $null \rightarrow (I_2:7) \rightarrow (I_1:4) \rightarrow (I_3:2) \rightarrow (I_5:1) \Rightarrow$ prefix $\{I_2, I_1, I_3\}$ count 1

$$\text{base} = \{ \{I_2, I_1\}:1, \{I_2, I_1, I_3\}:1 \}$$

$$I_2 = 1+1 = 2 \geq 2 \text{ keep} \quad I_1 = 1+1 = 2 \geq 2 \text{ keep} \quad I_3 = 1 < 2 \text{ pruned}$$

$$\text{conditional FP-tree} = \langle I_2:2, I_1:2 \rangle$$

$$\text{patterns: } \{I_2, I_5\}:2, \{I_1, I_5\}:2, \{I_2, I_1, I_5\}:2$$

- 8 **Suffix I_4 .** Two nodes.

Path 1: $null \rightarrow (I_2:7) \rightarrow (I_4:1) \Rightarrow$ prefix $\{I_2\}$ count 1

Path 2: $null \rightarrow (I_2:7) \rightarrow (I_1:4) \rightarrow (I_4:1) \Rightarrow$ prefix $\{I_2, I_1\}$ count 1

$$I_2 = 1+1 = 2 \geq 2 \text{ keep} \quad I_1 = 1 < 2 \text{ pruned}$$

$$\text{conditional FP-tree} = \langle I_2:2 \rangle \implies \{I_2, I_4\}:2$$

- 9 **Suffix I_3 .** Three nodes — this is the one that traps people.

Path 1: $null \rightarrow (I_2:7) \rightarrow (I_1:4) \rightarrow (I_3:2) \Rightarrow$ prefix $\{I_2, I_1\}$ count 2

Path 2: $null \rightarrow (I_2:7) \rightarrow (I_3:2) \Rightarrow$ prefix $\{I_2\}$ count 2

Path 3: $null \rightarrow (I_1:2) \rightarrow (I_3:2) \Rightarrow$ prefix $\{I_1\}$ count 2

$$I_2 = 2+2 = 4 \geq 2 \text{ keep} \quad I_1 = 2+2 = 4 \geq 2 \text{ keep}$$

$$\text{conditional FP-tree} = \langle I_2:4, I_1:2 \rangle, \langle I_1:2 \rangle \quad (\text{two branches})$$

I_2 and I_1 occur **together** only on Path 1, where the count is 2.

$$\text{patterns: } \{I_2, I_3\}:4, \{I_1, I_3\}:4, \{I_2, I_1, I_3\}:2$$

- 10 **Suffix I_1 .** Two nodes.

Path 1: $null \rightarrow (I_2:7) \rightarrow (I_1:4) \Rightarrow$ prefix $\{I_2\}$ count 4

Path 2: $null \rightarrow (I_1:2) \Rightarrow$ attached straight to null, empty prefix, contributes nothing

$$\text{conditional FP-tree} = \langle I_2:4 \rangle \implies \{I_2, I_1\}:4$$

- 11 **Summary table** — this is what the examiner wants to see.

SUFFIX	CONDITIONAL PATTERN BASE	CONDITIONAL FP-TREE	PATTERNS GENERATED
I5	$\{\{I_2, I_1\}:1, \{I_2, I_1, I_3\}:1\}$	$\langle I_2:2, I_1:2 \rangle$	$\{I_2, I_5\}:2, \{I_1, I_5\}:2, \{I_2, I_1, I_5\}:2$
I4	$\{\{I_2, I_1\}:1, \{I_2\}:1\}$	$\langle I_2:2 \rangle$	$\{I_2, I_4\}:2$
I3	$\{\{I_2, I_1\}:2, \{I_2\}:2, \{I_1\}:2\}$	$\langle I_2:4, I_1:2 \rangle, \langle I_1:2 \rangle$	$\{I_2, I_3\}:4, \{I_1, I_3\}:4, \{I_2, I_1, I_3\}:2$
I1	$\{\{I_2\}:4\}$	$\langle I_2:4 \rangle$	$\{I_2, I_1\}:4$

Why I_2 is never mined: it has the highest count so it sits directly under null — its conditional pattern base is empty. Every itemset containing I_2 was already produced under the other suffixes, and $\{I_2\}:7$ came from scan 1.

Answer 13 frequent itemsets — $\{I_2\}:7$, $\{I_1\}:6$, $\{I_3\}:6$, $\{I_4\}:2$, $\{I_5\}:2$; $\{I_2, I_1\}:4$, $\{I_2, I_3\}:4$, $\{I_1, I_3\}:4$, $\{I_2, I_4\}:2$, $\{I_2, I_5\}:2$, $\{I_1, I_5\}:2$; $\{I_2, I_1, I_3\}:2$, $\{I_2, I_1, I_5\}:2$.

Three common errors (1) Attaching T500/T700 to the existing I_1 instead of starting a new root branch — see step 5 of the construction. (2) Copying an ancestor's count into a prefix path instead of the suffix node's count. (3) Writing $\{I_2, I_1, I_3\}:4$ — the marginals are both 4, but the pair co-occurs on only one branch, so the answer is 2.

M23 ECLAT — vertical data format

Lecture 7B · 2023 Q5(a)

QUESTION

Consider the following transaction database of the “ABC” sales company. Using the ECLAT algorithm find the frequent pattern with $\text{min_sup} = 2$.

TID	LIST OF ITEM IDS	TID	LIST OF ITEM IDS	TID	LIST OF ITEM IDS
T100	I1, I2, I5	T400	I1, I2, I4	T700	I1, I3
T200	I2, I4	T500	I1, I3	T800	I1, I2, I3, I5
T300	I2, I3	T600	I2, I3	T900	I1, I2, I3

1 Key idea. Apriori and FP-growth use a **horizontal** format ($TID \rightarrow Items$); ECLAT uses a **vertical** one ($Itemset \rightarrow TID_set$). Then:

- support of an itemset = **length of its TID_set** ;
- support of a $(k+1)$ -itemset = length of the **intersection** of its k -itemsets' TID_sets .

After the single transforming scan, the database is never read again.

2 Transform to vertical format (one scan). List, for each item, the TIDs where it appears.

ITEMSET	TID_SET	SUPPORT	STATUS (≥ 2)
I1	{T100, T400, T500, T700, T800, T900}	6	Keep
I2	{T100, T200, T300, T400, T600, T800, T900}	7	Keep
I3	{T300, T500, T600, T700, T800, T900}	6	Keep
I4	{T200, T400}	2	Keep
I5	{T100, T800}	2	Keep

3 2-itemsets by intersection. Take each pair and keep only the TIDs common to both.

$$TID(I_1) \cap TID(I_2) = \{T100, T400, T800, T900\} \rightarrow 4 \geq 2 \text{ Keep}$$

$$TID(I_1) \cap TID(I_3) = \{T500, T700, T800, T900\} \rightarrow 4 \geq 2 \text{ Keep}$$

$$TID(I_1) \cap TID(I_4) = \{T400\} \rightarrow 1 < 2 \text{ Pruned}$$

$$TID(I_1) \cap TID(I_5) = \{T100, T800\} \rightarrow 2 \geq 2 \text{ Keep}$$

$$TID(I_2) \cap TID(I_3) = \{T300, T600, T800, T900\} \rightarrow 4 \geq 2 \text{ Keep}$$

$$TID(I_2) \cap TID(I_4) = \{T200, T400\} \rightarrow 2 \geq 2 \text{ Keep}$$

$$TID(I_2) \cap TID(I_5) = \{T100, T800\} \rightarrow 2 \geq 2 \text{ Keep}$$

$$TID(I_3) \cap TID(I_5) = \{T800\} \rightarrow 1 < 2 \text{ Pruned}$$

$\{I_3, I_4\}$ and $\{I_4, I_5\}$ give empty intersections and are also pruned.

$$L_2 = \{ \{I_1, I_2\}:4, \{I_1, I_3\}:4, \{I_1, I_5\}:2, \{I_2, I_3\}:4, \{I_2, I_4\}:2, \{I_2, I_5\}:2 \}$$

- 4 **3-itemsets** — intersect 2-itemsets that share a prefix.

$$TID(\{I_1, I_2\}) \cap TID(\{I_1, I_3\}) = \{T100, T400, T800, T900\} \cap \{T500, T700, T800, T900\} \\ = \{T800, T900\} \rightarrow 2 \text{ Keep}$$

$$TID(\{I_1, I_2\}) \cap TID(\{I_1, I_5\}) = \{T100, T400, T800, T900\} \cap \{T100, T800\} = \{T100, T800\} \rightarrow 2 \text{ Keep}$$

$$TID(\{I_1, I_3\}) \cap TID(\{I_1, I_5\}) = \{T500, T700, T800, T900\} \cap \{T100, T800\} = \{T800\} \rightarrow 1 \text{ Pruned}$$

$$L_3 = \{ \{I_1, I_2, I_3\}:2, \{I_1, I_2, I_5\}:2 \}$$

- 5 **4-itemset test and termination.**

$$TID(\{I_1, I_2, I_3\}) \cap TID(\{I_1, I_2, I_5\}) = \{T800, T900\} \cap \{T100, T800\} = \{T800\} \rightarrow 1 < 2 \text{ Pruned}$$

No 4-itemset satisfies *min_sup*; the algorithm terminates.

Answer L_1 : I1:6, I2:7, I3:6, I4:2, I5:2. L_2 : the six pairs above. L_3 : {I1,I2,I3}:2 and {I1,I2,I5}:2. Same 13 itemsets as FP-growth in M22.

Properties No database rescans after the transform; the *TID_set* carries all information needed for counting. **Weakness:** TID-sets get long and expensive to intersect — fixed by the **diffset** technique, which stores only $TID(X) \setminus TID(Y)$.

M24 Correlation analysis — lift

Lecture 7B

QUESTION

Of 10,000 transactions, 6,000 include computer games, 7,500 include videos and 4,000 include both. With *min_sup* = 30% and *min_conf* = 60%, evaluate the rule *buys*(*X*, “computer games”) $\Rightarrow$ *buys*(*X*, “videos”) and comment on its interestingness.

- 1 Support and confidence first.

$$support = \frac{4000}{10000} = 40\% \geq 30\% \quad confidence = \frac{4000}{6000} = 66\% \geq 60\%$$

Both thresholds are met, so the rule is **strong**.

- 2 But compare the confidence with the *unconditional* probability of buying videos.

$$P(\text{videos}) = \frac{7500}{10000} = 75\%$$

75% > 66%: knowing a customer bought computer games **decreases** the estimated chance they buy videos. The two are **negatively associated**, yet the rule passed both thresholds. Confidence alone can be deceiving — it estimates $P(B | A)$ but says nothing about the strength of the correlation.

- 3 Use **lift**. *A* is independent of *B* iff $P(A \cup B) = P(A)P(B)$; lift measures the departure from that.

$$Lift(A, B) = \frac{P(A \cup B)}{P(A)P(B)}$$

Lift > 1: positively correlated. *Lift* < 1: negatively correlated. *Lift* = 1: independent.

- 4 Substitute.

$$Lift(\text{game}, \text{video}) = \frac{0.40}{0.60 \times 0.75} = \frac{0.40}{0.45} = 0.89 < 1$$

Answer support 40%, confidence 66% — the rule is strong; but *Lift* = 0.89 < 1, so games and videos are **negatively correlated** and the rule is **not interesting**.

Fastest proof One line does it: $\text{confidence} = 66\% < P(\text{videos}) = 75\%$, therefore negative association.

Formula Sheet

ENTROPY

$$Info(D) = - \sum_{i=1}^m p_i \log_2 p_i$$

INFORMATION GAIN

$$Gain(A) = Info(D) - Info_A(D)$$

GAIN RATIO

$$GainRatio(A) = \frac{Gain(A)}{SplitInfo_A(D)}$$

GINI OF A BINARY SPLIT

$$Gini_A(D) = \frac{|D_1|}{|D|} Gini(D_1) + \frac{|D_2|}{|D|} Gini(D_2)$$

MIN / MAX CLUSTER DISTANCE

$$d_{min} = \min_{p \in C_i, p' \in C_j} |p - p'| \quad d_{max} = \max_{p \in C_i, p' \in C_j} |p - p'|$$

MERGES IN A DENDROGRAM

$$n - 1 \text{ levels for } n \text{ objects}$$

CENTROID / RADIUS

$$\vec{x}_0 = \frac{\vec{LS}}{N} \quad R = \sqrt{\frac{SS}{N} - \left(\frac{\vec{LS}}{N}\right)^2}$$

JACCARD / LINK

$$sim(T_i, T_j) = \frac{|T_i \cap T_j|}{|T_i \cup T_j|} \quad Link(T_i, T_j) = |N(T_i) \cap N(T_j)| \quad |N_\epsilon(p)| \geq MinPts, \quad N_\epsilon(p) = \{q : d(p, q) \leq \epsilon\}$$

SQUARED-DISTANCE SHORTCUT

$$d \leq \epsilon \iff \Delta x^2 + \Delta y^2 \leq \epsilon^2$$

BINARY DISSIMILARITY

$$d_{sym} = \frac{r + s}{q + r + s + t} \quad d_{asym} = \frac{r + s}{q + r + s}$$

ORDINAL MAPPING

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$

COSINE

$$\cos(d_1, d_2) = \frac{d_1 \bullet d_2}{\|d_1\| \|d_2\|}$$

RULES PER ITEMSET

$$2^k - 2$$

LIFT

$$Lift(A, B) = \frac{P(A \cup B)}{P(A)P(B)}$$

EXPECTED ENTROPY

$$Info_A(D) = \sum_{j=1}^v \frac{|D_j|}{|D|} Info(D_j)$$

SPLIT INFORMATION

$$SplitInfo_A(D) = - \sum_{j=1}^v \frac{|D_j|}{|D|} \log_2 \frac{|D_j|}{|D|}$$

GINI INDEX

$$Gini(D) = 1 - \sum_{i=1}^m p_i^2$$

NUMBER OF BINARY SPLITS

$$2^{k-1} - 1$$

MEAN / AVERAGE CLUSTER DISTANCE

$$d_{mean} = |m_i - m_j| \quad d_{avg} = \frac{1}{n_i n_j} \sum_{p \in C_i} \sum_{p' \in C_j} |p - p'|$$

CLUSTERING FEATURE

$$CF = \langle N, \vec{LS}, SS \rangle, \quad CF_3 = CF_1 + CF_2$$

DIAMETER

$$D = \sqrt{\frac{2N \cdot SS - 2\vec{LS}^2}{N(N-1)}}$$

DBSCAN CORE TEST

$$|N_\epsilon(p)| \geq MinPts, \quad N_\epsilon(p) = \{q : d(p, q) \leq \epsilon\}$$

NOMINAL DISSIMILARITY

$$d(i, j) = \frac{p - m}{p}$$

Z-SCORE

$$s_f = \frac{1}{n} \sum_i |x_{if} - m_f| \quad z_{if} = \frac{x_{if} - m_f}{s_f}$$

MINKOWSKI

$$d(i, j) = \sqrt[h]{\sum_f |x_{if} - x_{jf}|^h}$$

SUPPORT / CONFIDENCE

$$support = P(X \cup Y) \quad conf = \frac{\sigma(X \cup Y)}{\sigma(X)}$$

DHP HASH

$$H(x, y) = (order(x) \times 10 + order(y)) \bmod 7$$